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Tunable wave propagation and band gap characteristics in functionally graded lattices with multiple topologies via dynamic stiffness and Floquet–Bloch formulation

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Abstract Metamaterials’ design and development are a fast-growing field with unlimited potential in aircraft, acoustics, and telecommunications. Adding a gradient to materials could change wave behavior, making them useful for wave control. This study examines the impact of graded material distribution on dynamic wave transmission in 2D lattice structures. Three distinct lattice configurations, regular diamond, chiral, and triangular layouts, are used in the investigation. All lattices’ unit cell is represented using a Timoshenko beam with axial extension. Material inhomogeneity in the cell elements is represented through functionally graded properties, incorporated using three gradation schemes: Power law, exponential, and trigonometric distributions. The stiffness matrix is formulated using the dynamic stiffness method combined with the Floquet–Bloch theorem, and the resulting eigenvalue system is solved using the Wittrick–Williams algorithm. This investigation shows that modifying the material property along the thickness direction does not affect the shapes of the dispersion curves but uniformly scales the frequency magnitudes in all three lattice structures. Iso-frequency contours show that increased material inhomogeneity indicates enhanced anisotropy and direction-dependent wave propagation. The velocity charts illustrate that wave-propagation’s phase speed decreases across all lattices as index of material gradient increases. Validating all dispersion curves with finite element analysis verifies the results’ accuracy and reliability. This work highlights the use of functionally graded materials in lattice structures to tailor wave-propagation characteristics.

1 Introduction

The design and development of metamaterials have garnered significant attention due to their extraordinary capabilities in manipulating wave propagation beyond the limitations of non-gradient materials. Architected lattice materials (ALMs) are engineered structures known for their geometry-driven dynamic behaviors, including directional wave transmission, wave filtering, and acoustic cloaking capabilities. Among these ALMs, diamond [1, 2] chiral lattices [3, 4], triangular lattice [5, 6], hexagonal [7–9], re-entrant [10, 11], and star-shaped lattice [12, 13], configurations are notable for their distinctive geometric arrangements, which can lead to remarkable wave-propagation characteristics. Their periodicity allows for the formation of bandgaps, effectively blocking or guiding wave propagation in specific frequency ranges. Given ALMs’ unique properties, the study of wave propagation in lattice materials is crucial for engineering applications ranging from aerospace systems to multifunctional materials. A comprehensive understanding of their dynamic behavior under various loading conditions is vital for optimizing structural design.

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A considerable number of researchers have explored diverse lattice configurations, each with unique geometrical characteristics, to analyze and understand their wave-propagation behavior and associated dispersion phenomena. These investigations span a wide range of structural topologies and have contributed significantly to the development of engineered materials with tailored dynamic properties, particularly for applications in vibration attenuation, acoustic control, and mechanical metamaterials. For example, Mead [14] conducted foundational research on wave propagation in continuous periodic systems, emphasizing their relevance to real-world engineering structures. Building on this groundwork, later studies shifted focus toward the design and analysis of engineered periodic structures. Ruzzene [15] demonstrated the phenomenon of wave beaming in cellular materials, revealing how anisotropy can effectively channel energy along preferred directions. Phani [16] explored wave dispersion characteristics in two-dimensional periodic lattices, while Gonella [17] established a correlation between lattice geometry—including hexagonal and re-entrant patterns—and the dynamic behavior of graded lattice structures. Furthermore, Farzbod [18, 19] refined Bloch wave analysis techniques, offering improved methodologies for incorporating external forces in periodic systems. A noteworthy limitation of many of these studies, however, is the common assumption that the base materials used to construct these lattices are isotropic and homogeneous. This simplification often shifts the primary focus to topological optimization of the unit cell design, potentially overlooking the added degrees of freedom that can be harnessed through material composition engineering. Introducing material gradients or anisotropic properties could significantly expand the design space and enable the development of lattice structures with more versatile and tunable.

Recently, several researchers have started analyzing the effect of material property variations on the wave-propagation behavior of lattice structures by incorporating functionally graded materials (FGMs) [20–25]. This shift in focus highlights the potential of FGMs to introduce additional design flexibility and tunability in controlling wave characteristics within periodic systems mechanical responses. For example, Sepehri et al. [26] explored how functional gradation in planar metamaterials can be utilized to control elastic wave propagation, emphasizing the role of spatially varying material properties in tailoring dispersion behavior and tunable bandgaps. Jafari et al. [27] studied hybrid lattice metamaterials featuring auxiliary resonators made from functionally graded (FG) materials, showing how material gradation enhances wave manipulation and enables tunable bandgap control. Li et al. [28] demonstrated that inertial amplification in periodic functionally graded beams can induce multiple band gaps, significantly enhancing wave attenuation efficiency in composite structures across various frequencies. Li and Li [29] analyzed the vibro-acoustic response of functionally graded graphene origami auxetic plates using a refined plate theory combined with numerical simulation through finite element-based vibro-acoustic modeling, demonstrating enhanced sound transmission loss due to graded auxetic architecture. Although the above studies successfully use the finite element method (FEM) to analyze the wave propagation in FG lattice structures, FEM has certain limitations. It often requires a fine mesh to capture detailed features, leading to high computational costs, especially for large or high-frequency models. Additionally, FEM may sometimes lead to the formation of ill-conditioned matrices, which can affect numerical stability and accuracy. These challenges highlight the need for alternative methods that can provide more efficient and reliable analysis of FG lattice structures.

The dynamic stiffness method (DSM) [30–33], also known as the spectral element method (SEM), serves as a powerful alternative to the FEM. Unlike FEM, DSM offers higher accuracy with fewer degrees of freedom by formulating the exact dynamic stiffness (DS) matrix based on the governing differential equations. Some researchers have already applied this method to study FG lattice structures, demonstrating its effectiveness in capturing wave behavior and dispersion characteristics with reduced computational effort. For example, Wu et al. [34] used the SEM to analyze elastic wave propagation in 1D phononic crystal with functionally graded materials, examining band gap characteristics under various gradient profiles. Li et al. [35] explored elastic wave propagation in planar functionally graded micro-lattices using SEM and finite element modeling, showing how size effects and material gradation enable active band gap tuning. Yan et al. [36] investigated band gap characteristics in 2D functionally graded periodic grid structures using SEM, showing how material gradation and resonator design influence elastic wave attenuation behavior. Gao et al. [37] examined the free vibration and buckling behavior of FG graphene origami auxetic beams in thermal environments using higher-order shear deformation theory and analytical solution methods, showing strong sensitivity to both gradation and temperature effects. Dorn and Kochmann [38] formulated an analytical ray-theory framework to describe elastic wave propagation in graded metamaterials, revealing how spatial material gradients control wave trajectories and dispersion characteristics. Bacigalupo et al. [39] developed a dynamic homogenization framework for multi-layered lattice metamaterials with alternating chiral microstructures, employing analytical continualization and discrete-to-continuum modeling to predict dispersive wave behavior and tunable vibration-

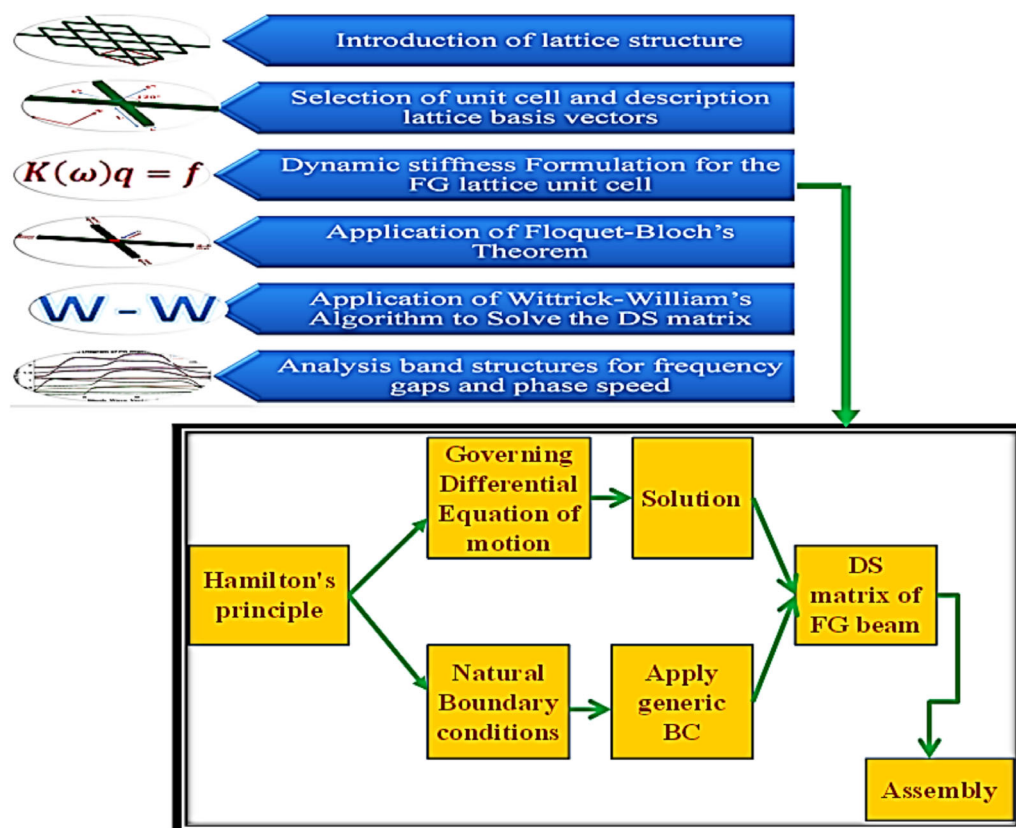


Fig. 1 Flowchart illustrating the methodological workflow for the investigation of wave behavior in two-dimensional FG lattice systems

filtering capabilities. One of the primary challenges in implementing the DSM lies in solving the frequency-dependent DS matrix. This challenge is effectively addressed by employing the Wittrick–Williams (W–W) algorithm [40], which facilitates accurate and efficient determination of natural frequencies and corresponding mode shapes. Mishra and Kumar [41] employed the dynamic stiffness method with the W–W algorithm to study FG square lattices. Their analysis is limited to power-law gradation and reported no band gaps within the first ten modes; a band gap emerged only after introducing geometric heterogeneity, occurring at a very high frequency of approximately 50,000 rad/s. Anisotropy and alternative lattice topologies were not examined.

However, it is important to note that existing DSM-based investigations predominantly address simple lattice configurations, such as one-dimensional or square lattices, where low-frequency bandgap formation is either absent or occurs only at very high frequencies following geometric heterogeneity. Moreover, these studies are largely restricted to power-law gradation and do not explore anisotropy or direction-dependent wave propagation. Consequently, the combined influence of lattice topology and alternative material gradation laws on wave-propagation characteristics remains insufficiently understood. To address this gap, the present study examines FG diamond, chiral, and triangular lattice configurations using the DSM in conjunction with the W–W algorithm, with particular emphasis on bandgap scaling and anisotropic wave behavior. This approach enables accurate and efficient computation of dispersion properties, offering new insights into the dynamic response of functionally graded lattice metamaterials.

To enhance clarity and maintain a logical progression, the structure of this paper follows a well-defined framework, as illustrated in the flowchart provided in Fig. 1. The paper begins with an introduction, followed by Sect. 2, which establishes the formulation of the DS matrix corresponding to unit cell configurations. This section also elaborates on the lattice geometry, material gradation, and notational framework. Section 3 focuses on the application of periodic boundary conditions and the employment of the W–W algorithm for determining eigenvalues. In Sect. 4, the dispersion characteristics of the proposed lattice systems are analyzed, with validation results provided for the diamond, chiral, and triangular configurations. The final section, Sect. 5, offers a concise overview of the key outcomes and their significance.

2 Dynamic stiffness formulation of FG 2D lattice structures

The unit cell approach based on Bloch's theorem is chosen to predict the wave-propagation characteristics of lattice systems made of FG material. This approach allows the prediction of wave behavior in the entire structure by examining just one unit cell. This simplifies the analysis and reduces computational time. Each unit cell is composed of Timoshenko beams connected in a specific manner. The DSM is used to analyze these beams and predict the wave transmission response of the lattice unit cell. A more thorough explanation of the DS matrix formulation procedure is provided below.

2.1 The notational conventions and description of lattice geometry

In this paper, three distinct lattice geometries are explored: (a) Diamond, (b) chiral, (c) triangular lattice, shown in Fig. 2a, c, and e, respectively. These lattice unit geometries are meticulously assembled to create entire periodic structures along specific directions. As shown in Fig. 2b, d, and f, these directions are called lattice basis vectors and are represented by the notation e_i , where $i = 1, 2$. Lattice points are the intersections of these unit lattice geometries.

It is widely accepted that studying wavevectors inside the first Brillouin zone is adequate to investigate the dispersion properties of the structured materials. Finding the material's frequency band gaps is frequently the main focus. The dispersion diagram's minimum and maximum points for each band define these band gaps. These extreme points are therefore especially significant. These extreme points are probably located on the edge of the irreducible Brillouin zone (IBZ) because most crystals and structured materials have inherent symmetry. Figure 3 shows the irreducible Brillouin zones (IBZs) for the diamond, square, and triangular lattices along with their respective high-symmetry point coordinates.

2.2 Material parameter representation for an FG beam

In this study, FG beams are strategically connected to form architected periodic lattice structures. The initial stage of the formulation, presented in Sect. 2.3, involves developing a dynamic stiffness model for an individual FG beam. This is followed by the construction of periodic lattices in Sect. 3, where Bloch's theorem is applied to investigate their wave-propagation behavior. As illustrated in Fig. 4a, the system consists of an FG beam of thickness h , length L , and width b , with material gradation introduced exclusively along the thickness direction. The material distribution is characterized using three gradation schemes—power law, exponential, and trigonometric—which define the continuous variation of Young's modulus E and density ρ through the thickness. The corresponding ceramic volume-fraction expressions for these gradation laws are provided in Eq. (1), while the resulting through-thickness variation of properties is shown in Fig. 4b–d.

$$\begin{aligned} \text{Power - law function} \quad V_c &= \left(\frac{z}{h} + \frac{1}{2}\right)^p, \quad -\frac{h}{2} \leq z \leq \frac{h}{2} \\ \text{Trigonometric function} \quad V_c &= \sin\left(\frac{\pi}{2}\left(\frac{z}{h}\right)^p\right), \quad 0 \leq z \leq h \\ \text{Exponential function} \quad V_c &= \frac{(e^{p\frac{z}{h}} - 1)}{(e^p - 1)}, \quad 0 \leq z \leq h \end{aligned} \quad (1)$$

In these formulations, z denotes the distance from the mid-plane (centroidal axis). The parameter p , referred to as the index of material gradient, governs the steepness of gradation for all three schemes—power law, exponential, and trigonometric. Under each model, the material distribution is defined such that the upper surface of the beam corresponds to the ceramic phase, while the lower surface corresponds to the metallic phase.

By applying the gradation functions defined in Eq. (1), the effective material properties of the FG beam—such as Young's modulus E and mass density ρ are obtained as:

$$\begin{aligned} \rho(z) &= \rho_b + (\rho_t - \rho_b)V_c \\ E(z) &= E_b + (E_t - E_b)V_c \end{aligned} \quad (2)$$

In this formulation, the subscripts b and t denote the bottom and top surfaces of the FG beam, respectively. In this study, the power-law index range $p = 0 - 10$ is adopted following established FGM practice [42, 43], representing realistic metal–ceramic gradation. Larger values yield marginal stiffness and bandgap variations, offering no additional physical insight.

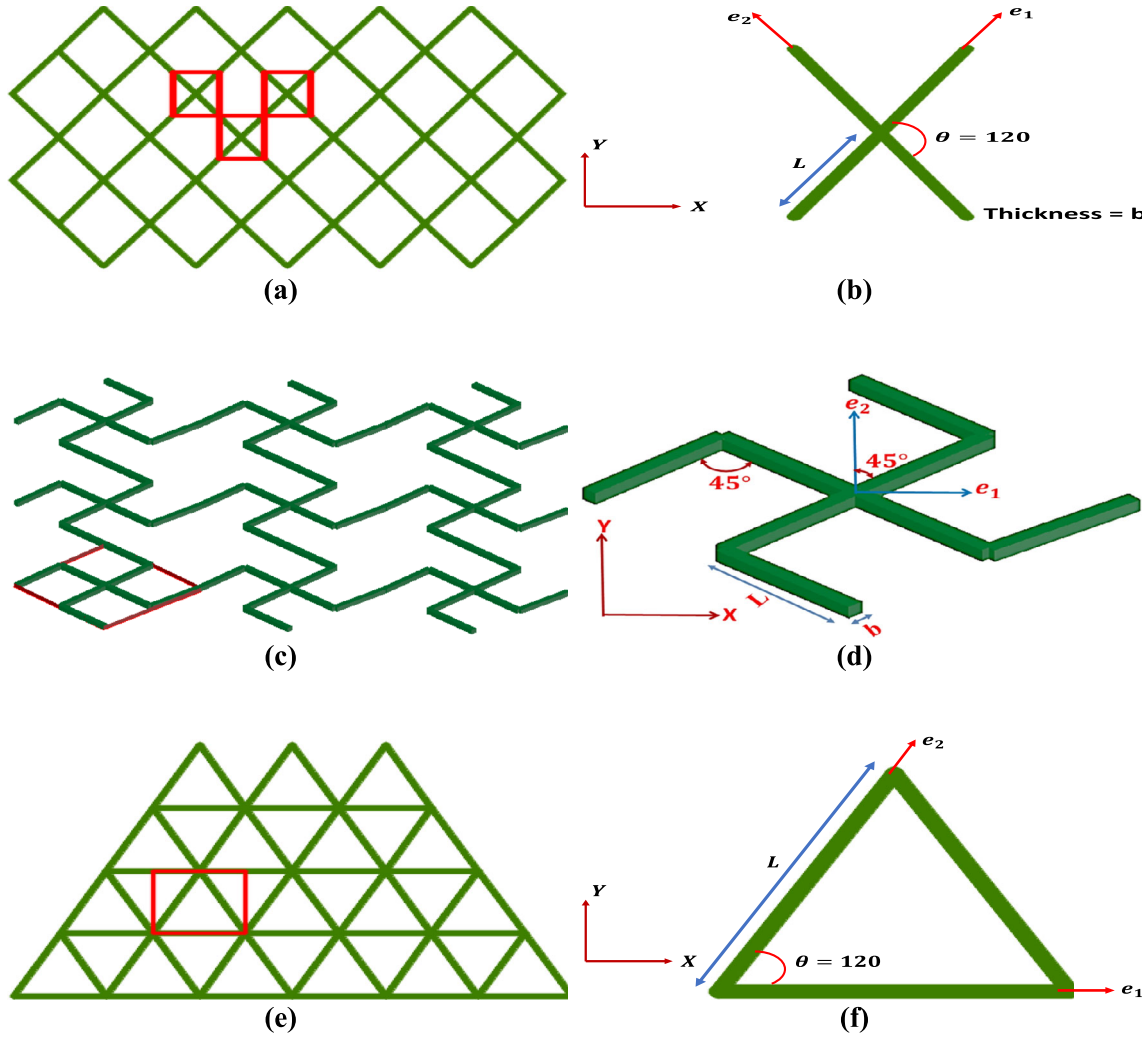


Fig. 2 The geometry of the three lattice structures and their corresponding unit cells. Subfigures **a**, **c**, and **e** present the overall configurations of the diamond, chiral, and triangular lattice structures, respectively. Subfigures **b**, **d**, and **f** show the detailed primitive unit cells for each structure, highlighting their geometric parameters. The basis vectors of the direct lattice, denoted as e_i

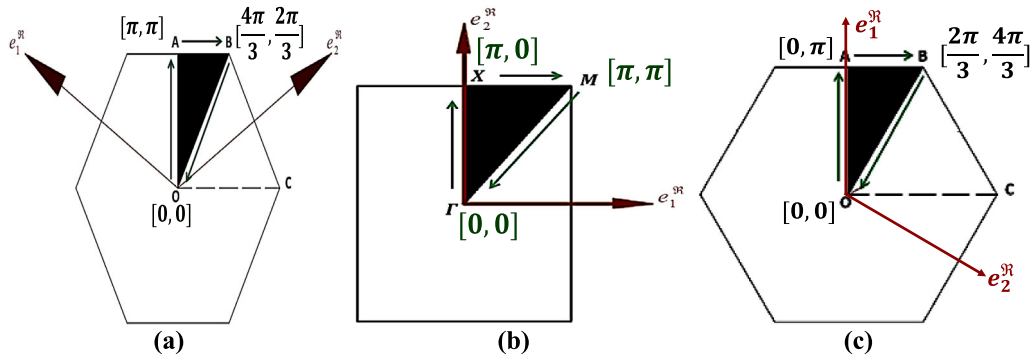


Fig. 3 The irreducible Brillouin zones (IBZ) along with the corresponding high-symmetry points for different lattice geometries: **a** Diamond lattice, **b** chiral lattice, and **c** triangular lattice. The coordinates of the high-symmetry points are indicated for each configuration, facilitating wavevector path definition for dispersion analysis

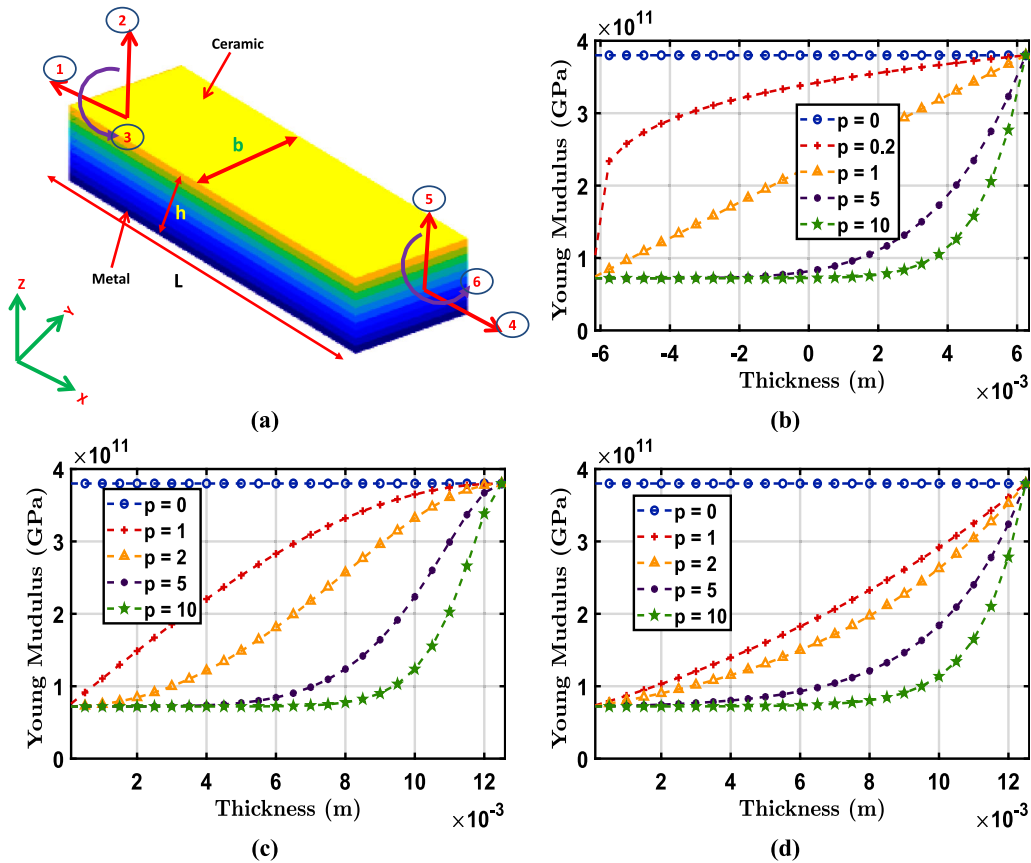


Fig. 4 a Schematic of an FG beam element with six degrees of freedom: Rotation about the y-axis (nodes 3 and 6), transverse displacement (nodes 2 and 5), and axial displacement (nodes 1 and 4). **b–d** Through-thickness distribution of Young's modulus corresponding to the power law, trigonometric, and exponential gradation schemes, respectively

2.3 Development of dynamic stiffness matrix for unit cell

This section summarizes the DSM formulation for FG lattice unit cells using Timoshenko beam theory (TBT) and lattice-based matrix assembly.

In this study, the displacement field of the Timoshenko beam theory is adopted to incorporate axial deformation, transverse motion, and cross-sectional rotation. For any point x on the beam, the axial and transverse displacements are defined as

$$\begin{aligned} u_a(x, t) &= \hat{u}(x, t) - z \left(\xi(x, t) + \frac{\partial w(x, t)}{\partial x} \right) \\ w_t(x, t) &= w(x, t) \end{aligned} \quad (3)$$

where $\hat{u}(x, t)$ denotes the mid-plane axial displacement, $w(x, t)$ is the transverse displacement, and $\xi(x, t)$ represents the rotation of the cross section. Utilizing the above displacement field, the Timoshenko beam theory (TBT) formulation for the free vibration of the FG beam is derived from the principle of virtual displacements and can be expressed as:

$$\begin{aligned} 0 &= \int_0^T \int_0^L [M_0 \left(\frac{\partial \hat{u}}{\partial t} \frac{\partial \delta \hat{u}}{\partial t} + \frac{\partial w}{\partial t} \frac{\partial \delta w}{\partial t} \right) \\ &\quad + M_2 \frac{\partial \xi}{\partial t} \frac{\partial \delta \xi}{\partial t} - N_{xx} \delta e_{xx}^{(0)} - M_{xx} \delta e_{xx}^{(1)} - Q_x \delta \gamma_{xz}] dx dt \end{aligned} \quad (4)$$

In this context, $\hat{u}$ and w denote the displacement components of the FG beam along the longitudinal (x) and transverse (y) directions, respectively, while ξ describes the rotation about the y -axis, and

$$\begin{aligned}\langle M_0, M_2 \rangle &= \int_{\hat{A}} \rho(z) \langle 1, (z - \check{n})^2 \rangle d\hat{A} \\ \langle E_0, E_2 \rangle &= \int_{\hat{A}} E(z) \langle 1, (z - \check{n})^2 \rangle d\hat{A}\end{aligned}\quad (5)$$

and

$$e_{xx}^{(0)} = \frac{\partial \hat{u}}{\partial x}, \quad e_{xx}^{(1)} = \frac{\partial \xi}{\partial x}, \quad \gamma_{xz} = \xi + \frac{\partial w}{\partial x} \quad (6)$$

and

$$\begin{aligned}N_{xx} &= \int_{\hat{A}} E(z) \frac{\partial \hat{u}}{\partial x} d\hat{A} \\ M_{xx} &= \int_{\hat{A}} E(z) (z - \check{n})^2 \frac{\partial \xi}{\partial x} d\hat{A} \\ Q_x &= K_s \int_{\hat{A}} \frac{E(z)}{(1 + \nu)} \left(\xi + \frac{\partial w}{\partial x} \right) d\hat{A}\end{aligned}\quad (7)$$

where $\check{n}$ corresponds to the true neutral surface in the context of the FG beam, whose expression is presented in the Appendix. Here, ν is the Poisson's ratio, $\hat{A}$ is the area of the longitudinal cross section of the beam, and K_s is the shear correction factor.

From Eq. (4), the corresponding Euler–Lagrange equations are obtained and presented in Eq. (8). The detailed derivation leading to Eq. (8) is provided in Appendices A and B.

$$\begin{aligned}\frac{\partial N_{xx}}{\partial x} &= M_0 \frac{\partial^2 \hat{u}}{\partial t^2}; \\ \frac{\partial Q_x}{\partial x} &= M_0 \frac{\partial^2 w}{\partial t^2}; \\ \frac{\partial M_{xx}}{\partial x} - Q_x &= M_2 \frac{\partial^2 \xi}{\partial t^2}\end{aligned}\quad (8)$$

The equilibrium equation for the FG Timoshenko beam is established through the substitution of Eq. (7) into Eq. (8), followed by simplification

$$\begin{aligned}\frac{\partial}{\partial x} \left[\left(\int_{\hat{A}} E(z) d\hat{A} \right) \left[\frac{\partial \hat{u}}{\partial x} \right] \right] &= M_0 \frac{\partial^2 \hat{u}}{\partial t^2}; \\ \frac{\partial}{\partial x} \left[K_s \left(\int_{\hat{A}} \frac{E(z)}{(1 + \nu)} d\hat{A} \right) \left(\xi + \frac{\partial w}{\partial x} \right) \right] &= M_0 \frac{\partial^2 w}{\partial t^2}; \\ \frac{\partial}{\partial x} \left\{ \left(\int_{\hat{A}} E(z) (z - \check{n})^2 d\hat{A} \right) \frac{\partial \xi}{\partial x} \right\} - \left[K_s \left(\int_{\hat{A}} \frac{E(z)}{(1 + \nu)} d\hat{A} \right) \left(\xi + \frac{\partial w}{\partial x} \right) \right] &= M_2 \frac{\partial^2 \xi}{\partial t^2}\end{aligned}\quad (9)$$

The solution of above partial differential Eq. (9) can be assumed as

$$\begin{aligned}\hat{u}(x, t) &= \bigcup''(x) e^{i\omega t} \\ w(x, t) &= w(x) e^{i\omega t} \\ \xi(x, t) &= \xi(x) e^{i\omega t}\end{aligned}\quad (10)$$

The functions $\bigcup''(x)$, $w(x)$, and $\xi(x)$ correspond to the vibration amplitudes associated with $\hat{u}$, w , and ξ , and ω denotes the natural angular frequency characterizing the free vibration of the FG beam.

The DS matrix for the FG Timoshenko beam is obtained by substituting Eq. (10) into Eq. (9) and by applying the geometric and force boundary conditions shown in Fig. 5. Substituting the displacement field $\hat{u}$

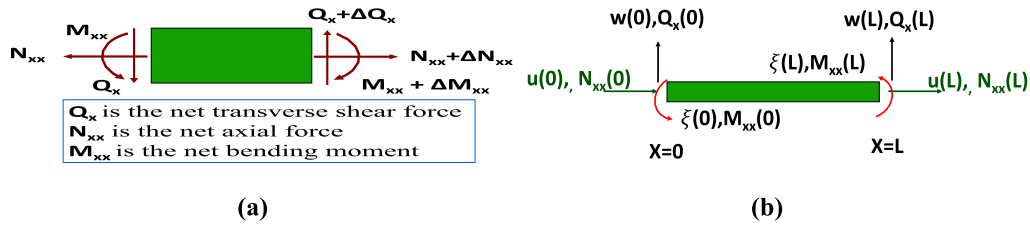


Fig. 5 **a** Depiction of internal force equilibrium conditions in an FG beam element **b** boundary condition (displacement and load) imposed on an FG beam element

into Eq. (9) and carrying out the complete algebraic simplifications and manipulations yields the following Eq. (11):

$$\begin{aligned}
 \frac{d}{dx} \left[\left(\int_{\hat{A}} E(z) d\hat{A} \right) \left[\frac{\partial \hat{u}}{\partial x} \right] \right] &= M_0 \frac{\partial^2 \hat{u}}{\partial t^2} \\
 \Rightarrow -\frac{d}{dt} \left(M_0 \frac{\partial \hat{u}(x) e^{i\omega t}}{\partial t} \right) + \frac{d}{dx} \left(E_0 \frac{\partial \hat{u}(x) e^{i\omega t}}{\partial x} \right) &= 0 \\
 \Rightarrow -\frac{d}{dt} \left(M_0 i \omega \hat{u}(x) e^{i\omega t} \right) + \frac{d}{dx} \left(E_0 e^{i\omega t} \frac{d \hat{u}(x)}{dx} \right) &= 0 \\
 \Rightarrow \left(M_0 \omega^2 \hat{u}(x) + E_0 \frac{d^2 \hat{u}(x)}{dx^2} \right) e^{i\omega t} &= 0 \\
 \Rightarrow M_0 \omega^2 \hat{u}(x) + E_0 \frac{d^2 \hat{u}(x)}{dx^2} &= 0
 \end{aligned} \tag{11}$$

The corresponding expression for the amplitude of the longitudinal force is given by:

$$\begin{aligned}
 N_A(x, t) &= -E_0 \frac{\partial \hat{u}(x, t)}{\partial x} = -E_0 \frac{\partial \hat{u}(x) e^{i\omega t}}{\partial x} \\
 \Rightarrow N_A(x, t) &= N_{XX}(x) e^{i\omega t} \\
 \text{where, } N_{XX}(x) &= -E_0 \frac{d \hat{u}(x)}{dx}
 \end{aligned} \tag{12}$$

Utilizing the differential operator $D = \frac{d}{dx}$, Eq. (11) takes the following form:

$$\begin{aligned}
 M_0 \omega^2 \hat{u}(x) + E_0 \frac{d^2 \hat{u}(x)}{dx^2} &= 0 \\
 \Rightarrow (E_0 D^2 + M_0 \omega^2) \hat{u}(x) &= 0 \\
 \Rightarrow (D^2 + \lambda_a^2) \hat{u}(x) &= 0 \\
 \text{where, } \lambda_a &= \sqrt{\frac{M_0 \omega^2}{E_0}}
 \end{aligned} \tag{13}$$

The general solution of Eq. (11) may be formulated as:

$$\hat{u}(x) = a_1 \sin(\lambda_a x) + a_2 \cos(\lambda_a x) \tag{14}$$

The substitution of Eq. (14) into Eq. (12) leads to the following formulation of the axial force $N_{XX}(x)$:

$$\begin{aligned} N_{XX}(x) &= -E_0 \frac{d^2 U(x)}{dx^2} = -E_0 \frac{d(a_1 \sin(\bar{\lambda}_a x) + a_2 \cos(\bar{\lambda}_a x))}{dx} \\ &\Rightarrow N_{XX}(x) = -E_0 \bar{\lambda}_a (a_1 \cos(\bar{\lambda}_a x) - a_2 \sin(\bar{\lambda}_a x)) \end{aligned} \quad (15)$$

In an analogous manner, the substitution of w and ξ into Eq. (9), followed by the required algebraic reductions, produces the following expression:

$$\begin{aligned} \frac{d}{dt} \left(\left(\int_{\hat{A}} E(z) d\hat{A} \right) \frac{\partial w(x) e^{i\omega t}}{\partial t} \right) - \frac{d}{dx} \left(K_s \left(\int_A \frac{E(z)}{(1+\nu)} d\hat{A} \right)_e \left(\frac{\partial w(x) e^{i\omega t}}{\partial x} - \xi(x) e^{i\omega t} \right) \right) &= 0 \\ \Rightarrow \frac{d}{dt} (E_0 i \omega w(x) e^{i\omega t}) - K_s G_0 \frac{d}{dx} \left(\frac{\partial w(x)}{\partial x} - \xi(x) \right) e^{i\omega t} &= 0 \\ \Rightarrow E_0 (i\omega)^2 w(x) e^{i\omega t} - K_s G_0 \frac{d}{dx} \left(\frac{\partial w(x)}{\partial x} - \xi(x) \right) e^{i\omega t} &= 0 \\ \Rightarrow E_0 \omega^2 w(x) + K_s G_0 \frac{d}{dx} \left(\frac{\partial w(x)}{\partial x} - \xi(x) \right) &= 0 \\ \text{where } G_0 &= \frac{E_0}{(1+\nu)} \end{aligned} \quad (16)$$

and

$$\begin{aligned} \frac{d}{dt} \left(M_2 \frac{\partial \xi(x) e^{i\omega t}}{\partial t} \right) - \frac{d}{dx} \left(E_2 \frac{\partial \xi(x) e^{i\omega t}}{\partial x} \right) - K_s G_0 \left(\frac{\partial w(x) e^{i\omega t}}{\partial x} - \xi(x) e^{i\omega t} \right) &= 0 \\ \Rightarrow \frac{d}{dt} (M_2 i \omega \xi(x) e^{i\omega t}) - \frac{d}{dx} (E_2 e^{i\omega t} \frac{d\xi(x)}{dx}) - K_s G_0 \left(\frac{dw(x)}{dx} e^{i\omega t} - \xi(x) e^{i\omega t} \right) &= 0 \\ \Rightarrow M_2 (i\omega)^2 \xi(x) e^{i\omega t} - E_2 e^{i\omega t} \frac{d^2 \xi(x)}{dx^2} - K_s G_0 \left(\frac{dw(x)}{dx} - \xi(x) \right) e^{i\omega t} &= 0 \\ \Rightarrow M_2 \omega^2 \xi(x) + E_2 \frac{d^2 \xi(x)}{dx^2} + K_s G_0 \left(\frac{dw(x)}{dx} - \xi(x) \right) &= 0 \end{aligned} \quad (17)$$

Based on the above expressions, the amplitudes of the shear force (Q_s) and bending moment (M_s) can be written as:

$$\begin{aligned} \text{Shear Force : } Q_s(x, t) &= K_s G_0 \left(\frac{\partial w(x, t)}{\partial x} - \xi(x, t) \right) = K_s G_0 \left(\frac{\partial w(x) e^{i\omega t}}{\partial x} - \xi(x) e^{i\omega t} \right) \\ &\Rightarrow Q_s(x, t) = K_s G_0 \left(\frac{\partial w(x)}{\partial x} - \xi(x) \right) e^{i\omega t} \\ \text{Shear Force Amplitude : } Q_X(x) &= K_s G_0 \left(\frac{\partial w(x)}{\partial x} - \xi(x) \right) \\ \text{Bending Moment : } M_s(x, t) &= E_2 \frac{\partial \xi(x, t)}{\partial x} = E_2 \frac{\partial \xi(x) e^{i\omega t}}{\partial x} \\ &\Rightarrow M_s(x, t) = E_2 \frac{\partial \xi(x)}{\partial x} e^{i\omega t} \\ \text{Bending Moment Amplitude : } M_{XX}(x) &= E_2 \frac{\partial \xi(x)}{\partial x} \end{aligned} \quad (18)$$

The coupling of Eqs. (16) and (17) yields a fourth-order ordinary differential equation, which holds for both $w(x)$ and $\xi(x)$, as shown below:

$$\begin{aligned} \{E_2 G_0 K_s D^4 + \omega^2 (E_0 E_2 + K_s G_0 M_2) D^2 + E_0 \omega^2 (M_2 \omega^2 - G_0 K_s)\} Z &= 0 \\ \{D^4 + \bar{b}^2 (r^2 + s^2) D^2 + \bar{b}^2 (1 - \bar{b}^2 r^2 s^2)\} Z &= 0 \\ \text{where } D &= \frac{d}{dx}, \quad Z = w(x) \text{ or } \xi(x) \\ \text{and } \bar{b}^2 &= \frac{E_0 \omega^2}{E_2}, r^2 = \frac{E_2}{G_0 K_s}, s^2 = \frac{M_2}{E_0} \end{aligned} \quad (19)$$

By adopting a trial solution with an undetermined constant λ , the auxiliary equation associated with Eq. (19) becomes:

$$\begin{aligned} \lambda^4 + \bar{b}^2(r^2 + s^2)\lambda^2 + \bar{b}^2(1 - \bar{b}^2 r^2 s^2) &= 0 \\ \left. \begin{matrix} \lambda_1 \\ \lambda_2 \end{matrix} \right\} &= \bar{b} \left\{ \pm \Delta/2 + \left[\Delta^2/4 + (1 - \bar{b}^2 r^2 s^2)/\bar{b}^2 \right]^{1/2} \right\}^{1/2} \\ \text{where } \Delta &= r^2 + s^2 \end{aligned} \quad (20)$$

Accordingly, for the two characteristic roots obtained from Eq. (19), the corresponding expressions for the bending displacement w rotation ξ can be written as:

$$\begin{aligned} w(x) &= A_1 \cosh(\lambda_1 x) + A_2 \sinh(\lambda_1 x) + A_3 \cosh(\lambda_2 x) + A_4 \sinh(\lambda_2 x) \\ \xi(x) &= B_1 \cosh(\lambda_1 x) + B_2 \sinh(\lambda_1 x) + B_3 \cosh(\lambda_2 x) + B_4 \sinh(\lambda_2 x) \end{aligned} \quad (21)$$

Substitution of the preceding solution into Eq. (16) yields the set of relations linking B_1 – B_4 with A_1 – A_4 .

$$\begin{aligned} B_1 &= k_1 A_2, \quad B_2 = -k_1 A_1 \\ B_3 &= k_2 A_4, \quad B_4 = -k_2 A_3 \\ \text{where} \\ k_1 &= (\lambda_1^2 - \bar{b}^2 s^2)/\lambda_1, \quad k_2 = (\lambda_2^2 + \bar{b}^2 s^2)/\lambda_2 \end{aligned} \quad (22)$$

With these coefficient relationships in place, the displacement fields and the corresponding forces and moments for the beam's transverse vibration take the following form:

$$\begin{aligned} w(x) &= A_1 \cosh(\lambda_1 x) + A_2 \sinh(\lambda_1 x) + A_3 \cosh(\lambda_2 x) + A_4 \sinh(\lambda_2 x) \\ \xi(x) &= k_1 A_2 \cosh(\lambda_1 x) - k_1 A_1 \sinh(\lambda_1 x) + k_2 A_4 \cosh(\lambda_2 x) - k_2 A_3 \sinh(\lambda_2 x) \\ Q_X(x) &= K_s G_0 \left(\frac{dw(x)}{dx} - \xi(x) \right) \\ \Rightarrow Q_X(x) &= K_s G_0 \left(\frac{d(A_1 \cosh(\lambda_1 x) + A_2 \sinh(\lambda_1 x) + A_3 \cosh(\lambda_2 x) + A_4 \sinh(\lambda_2 x))}{dx} \right. \\ &\quad \left. - k_1 A_2 \cosh(\lambda_1 x) + k_1 A_1 \sinh(\lambda_1 x) - k_2 A_4 \cosh(\lambda_2 x) + k_2 A_3 \sinh(\lambda_2 x) \right) \\ \Rightarrow Q_X(x) &= K_s G_0 \left[(\lambda_1 - k_1) \sinh(\lambda_1 x) A_1 + (\lambda_1 + k_1) \cosh(\lambda_1 x) A_2 + \right. \\ &\quad \left. (\lambda_2 - k_2) \sinh(\lambda_2 x) A_3 + (\lambda_2 + k_2) \cosh(\lambda_1 x) A_4 \right] \\ M_{XX}(x) &= E_2 \frac{d\xi(x)}{dx} \\ \Rightarrow M_{XX}(x) &= E_2 \frac{d(k_1 A_2 \cosh(\lambda_1 x) - k_1 A_1 \sinh(\lambda_1 x) + k_2 A_4 \cosh(\lambda_2 x) - k_2 A_3 \sinh(\lambda_2 x))}{dx} \\ \Rightarrow M_{XX}(x) &= E_2 \left[k_1 A_2 \lambda_1 \sinh(\lambda_1 x) - k_1 A_1 \lambda_1 \cosh(\lambda_1 x) \right. \\ &\quad \left. + k_2 A_4 \lambda_2 \sinh(\lambda_2 x) - k_2 A_3 \lambda_2 \cosh(\lambda_2 x) \right] \end{aligned} \quad (23)$$

Applying the geometric and force boundary conditions of Fig. 5 to Eqs. (14), (15), and (23), upon eliminating all the constants, yields the displacement–force relationship for the FG beam, expressed through the dynamic stiffness (DS) matrix as:

$$\mathbf{K}^{\ddot{\mathbf{E}}}(\omega) \hat{\mathbf{q}} = \mathbf{F} \quad (24)$$

where $\hat{q} = \left[\begin{matrix} \ddot{u}(0) & w(0) & \xi(0) & \ddot{u}(L) & w(L) & \xi(0) \end{matrix} \right]^T$ is nodal displacement vector, $F = \left[\begin{matrix} N_{XX}(0) & Q_X(0) & M_{XX}(0) & N_{XX}(L) & Q_X(L) & M_{XX}(L) \end{matrix} \right]^T$ is force and moment vector of FG beam and $K^E(\omega)$ is the elemental DS matrix containing transcendental function of ω , the circular natural frequency.

$$K^E(\omega) = \begin{bmatrix} a_1 & 0 & 0 & a_2 & 0 & 0 \\ 0 & b_{11} & b_{12} & 0 & b_{13} & b_{14} \\ 0 & b_{12} & b_{22} & 0 & -b_{14} & b_{24} \\ a_2 & 0 & 0 & a_1 & 0 & 0 \\ 0 & b_{13} & -b_{14} & 0 & b_{11} & -b_{12} \\ 0 & b_{14} & b_{24} & 0 & -b_{12} & b_{22} \end{bmatrix} \quad (25)$$

In the above DS matrix, the entries denoted by the letter ‘a’ represent axial stiffness, while those denoted by the letter ‘b’ correspond to bending stiffness. The complete expressions for these entries are provided in the Appendix C.

Once the DS matrix for the FG beam element is formulated, Eq. (26) is used to convert it to the global coordinate framework.

$$K^E(\omega, \varphi) = T_F^T(\varphi) K^E(\omega) T_F(\varphi) \quad (26)$$

where K^E is the individual element DS matrix in a global coordinate system, T_F represents the corresponding transformation matrix, and φ represents the angular orientation of the FG beam element relative to the global coordinate system.

$$T_F(\varphi) = \begin{bmatrix} T_1 & o_1 \\ o_1 & T_1 \end{bmatrix} \quad (27)$$

where o_1 is zero matrix of size 3×3 and

$$T_1 = \begin{bmatrix} \cos[\varphi] & \sin[\varphi] & 0 \\ -\sin[\varphi] & \cos[\varphi] & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (28)$$

For the three FG lattice unit cells, the global DS matrix $K^O(\omega)$ is derived through the integration of the DS matrices of their constituent beam elements, following the orientation of the lattice basis vectors. This assembly process is analogous to the standard procedure used in the FEM. For example, the assembly process for a diamond lattice consisting of four beam elements follows the same methodology as described in the Eq. (29)

$$K^O(\omega) = K^E\check{G}\left(\omega, \frac{\pi}{3}\right) + K^E\check{G}\left(\omega, \frac{\pi}{3}\right) + K^E\check{G}\left(\omega, \frac{2\pi}{3}\right) + K^E\check{G}\left(\omega, \frac{2\pi}{3}\right) \quad (29)$$

Lastly, the Floquet–Bloch [2] theorem is used to apply periodic boundary conditions (PBCs) to the global dynamic stiffness matrix of each of the three FG lattice structures’ unit cells. The process for doing so is described in Section 3.

3 Periodic boundary condition

This section describes the application of PBCs to the global DS matrix ($K^O(\omega)$) of the FG-DL system using the Floquet–Bloch theorem. As illustrated in Fig. 6, the diamond lattice comprises four Timoshenko beam elements and five nodes—one internal and four external. Input nodes (q_1 and q_3) and output nodes (q_2 and q_4) are further subdivided from the external nodes. The displacement fields at the output nodes may be represented in terms of the input node displacements using the Floquet–Bloch formulation as

$$\begin{aligned} q_2 &= e^{-i\beta_1} q_1 \\ q_4 &= e^{-i\beta_2} q_3 \end{aligned} \quad (30)$$

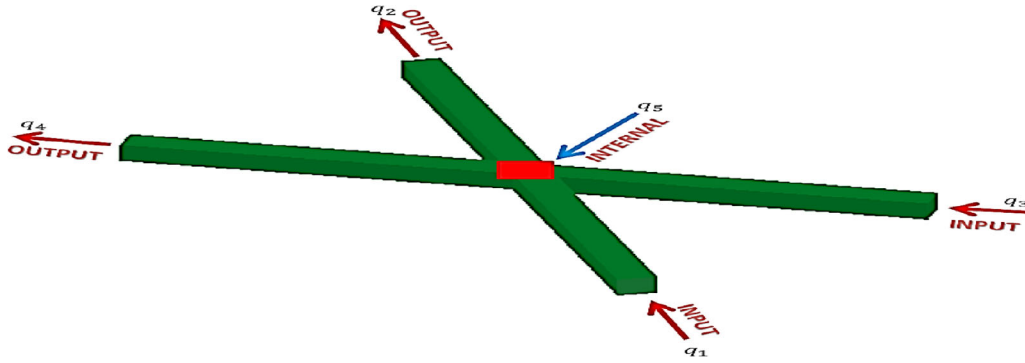


Fig. 6 Diamond unit cell of periodic structure in 2D space illustrating input, internal, and output nodes

where the wavenumbers corresponding to the reciprocal lattice directions are denoted by β_1 and β_2 . The generalized displacement vector of the FG diamond unit cell may be expressed using Eq. (30) as

$$q_u = T_b(\beta_1, \beta_2) \tilde{q}_{ui} \quad (31)$$

where $\tilde{q}_{ui} = [q_1, q_3, q_5]$ denote the reduced displacement vector representing the independent degrees of freedom within the unit cell, while $q_u = [q_1, q_2, q_3, q_4, q_5]$ represents the complete displacement vector. The transformation matrix $T_b(\beta_1, \beta_2)$, which accounts for periodic boundary conditions, is defined as follows

$$T_b(\beta_1, \beta_2) = \begin{bmatrix} I & 0 & 0 \\ e^{-i\beta_1} I & 0 & 0 \\ 0 & I & 0 \\ 0 & e^{-i\beta_2} I & 0 \\ 0 & 0 & I \end{bmatrix} \quad (32)$$

In this case, I denotes a 3×3 unitary matrix.

After applying the PBCs, the overall dynamic stiffness (DS) matrix is obtained as

$$K^O(\omega, \beta_1, \beta_2) = T_b^H(\beta_1, \beta_2) \cdot K^O(\omega) \cdot T_b(\beta_1, \beta_2) \quad (33)$$

In this context, $T_b^H(\beta_1, \beta_2)$ represents the Hermitian (conjugate transpose) of the transformation matrix $T_b(\beta_1, \beta_2)$.

3.1 Wittrick–William algorithm

With transcendental functions of ω , ε_1 , and ε_2 , the overall DS matrix $K^O(\omega, \varepsilon_1, \varepsilon_2)$ is extremely complicated and presents a difficult nonlinear eigenvalue issue. Researchers frequently employ the Wittrick–Williams method, a smart technique that uses Sturm sequences, to address issue. This algorithm's remarkable efficiency and great accuracy allow it to solve these complex problems quickly, saving a significant amount of time. The algorithm's application procedure is shown in Fig. 7.

4 Results and discussion

Wave-propagation characteristics of all three lattice configurations are investigated in this section using the DSM. Initially, the dynamic stiffness matrix of the lattice unit cell is formulated using a custom-developed Mathematica script. The resulting matrix is then exported to MATLAB using the “ToMatlab” function for further numerical processing and analysis. A detailed implementation of this workflow has been demonstrated previously for square lattice configurations [40]. The properties related to the geometry and material composition of the FGM beam is summarized in Table 1.

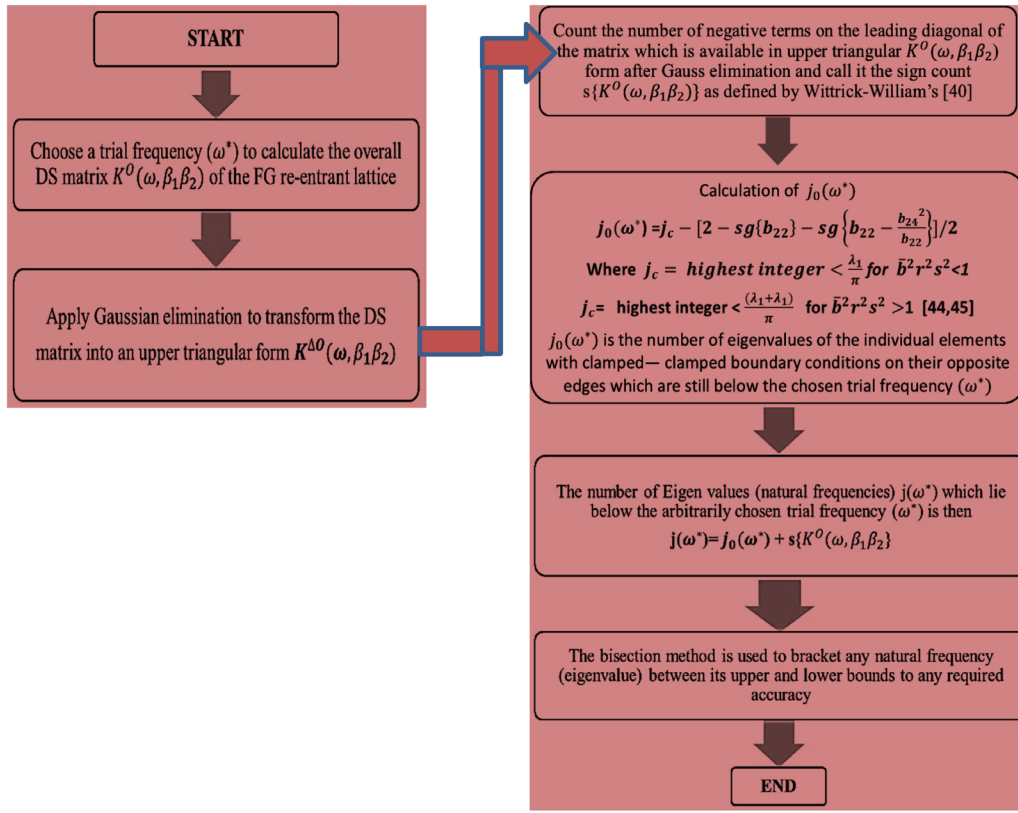


Fig. 7 Schematic representation of the procedure followed in applying the Wittrick–Williams algorithm (The terms appeared in the calculation of $j_0(\omega^*)$ [44, 45] are presented in the Appendix C)

Table 1 Specification of geometric dimensions and material properties for FG beam

Material parameters			
	Modulus of elasticity	Density	Poisson's ratio
Aluminum (bottom surface)	72 GPa	2702 kg/m ³	0.3
Alumina (top surface)	380 GPa	3960 kg/m ³	0.3
Geometric parameter			
	Length	Depth	Width
	0.125 m	0.0125 m	0.0125 m

4.1 Band structure validation of the FG diamond, chiral, and triangular lattices

To validate both the MATLAB code implementation and the DSM formulation for the selected FG diamond, FG chiral, and FG triangular lattice structures, benchmark data from existing literature on their isotropic counterparts were utilized. For consistency and direct comparison, validation is conducted using an index of material gradient of $p = 0$, representing a fully ceramic configuration. Figure 8a compares the dispersion curves of the diamond lattice with those reported by Leamy [2]; Fig. 8b presents the validation results for the chiral lattice against the findings of Kai Zhang [44]; and Fig. 8c compares the triangular lattice outcomes with those reported by Phani [16]. The close agreement in all cases confirms the accuracy and robustness of the proposed DSM framework and the associated MATLAB implementation.

As an alternative validation approach, the dispersion characteristics of the diamond lattice structure are examined through a comparative analysis to assess the accuracy of the MATLAB code implementation and the DSM formulation. FEM and COMSOL Multiphysics tools are also used to compute the FG-DL dispersion curve. In FEM approach, each structural member of the FG-DL unit cell is discretized into 30 finite elements, leading to a total of 120 elements for the full unit cell representation. The unit cell of the FG-DL is represented

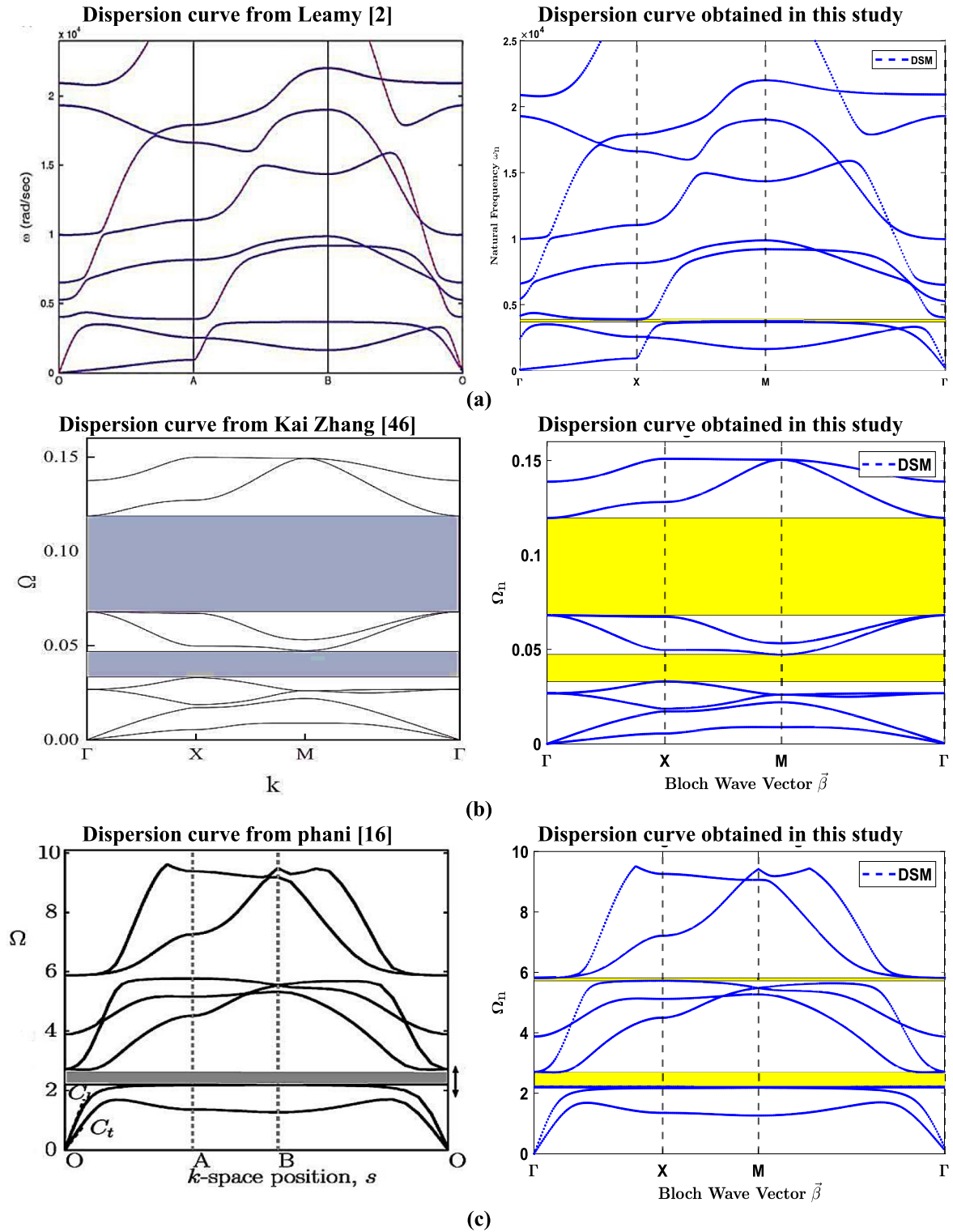


Fig. 8 Validation of dispersion curves for different lattice configurations: **a** Diamond lattice: Comparison between the results from Leamy [2] with $h = L/10$ (left) and the present study with $h = L/15$ (right). **b** Chiral lattice: Comparison between Kai Zhang [46] with $h = L/15$ (left) and the present study with the same configuration (right). **c** Triangular lattice: Comparison between Phani [16] with $h = L/50$ (left) and the results obtained in this study for identical geometry (right)

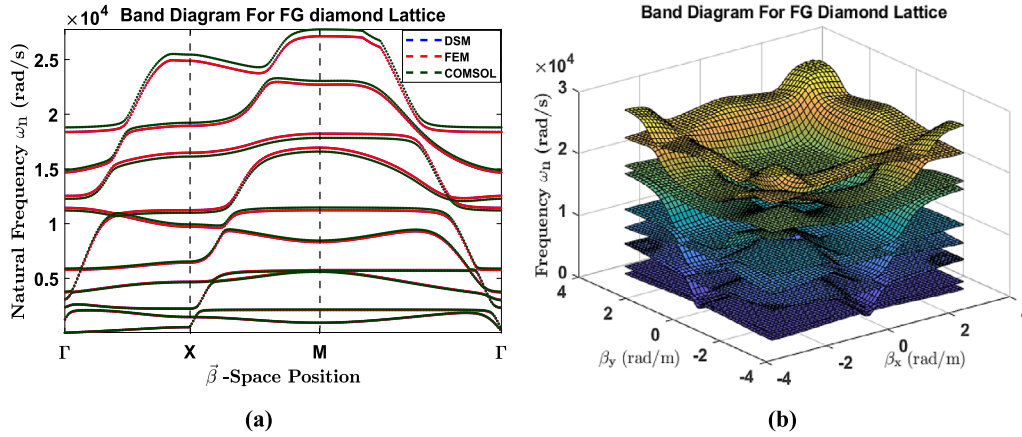


Fig. 9 Comparison of the first ten dispersion curves of the FG-DL structure calculated through the DSM, FEM, and COMSOL (with $p = 5$ and $h = L/10$)

by a solid model with four components in total and fine meshing in the COMSOL Multiphysics software model. Figure 9 displays the outcomes of the three approaches.

As observed in Fig. 9, all three methods show close agreement in the lower modes. However, for higher modes, FEM approach consistently yields less accurate results compared to the DSM method, with the corresponding frequency magnitudes from FEM being noticeably higher than those obtained using DSM. This discrepancy can be attributed to numerical errors arising from the discretization process in the FEM analysis. In contrast to FEM, some dispersion modes computed using COMSOL deviate both above and below the DSM results. These discrepancies can be attributed to a combination of finite element discretization effects and the modeling differences between COMSOL's solid element formulation and the beam-based approach used in DSM. Furthermore, the geometrical neutral surface is taken into account in the COMSOL solid model, whereas the physical neutral surface is employed in the DSM and FEM approach. These variations might contribute to greater errors observed in the COMSOL simulations.

Table 2 presents the first ten eigenfrequencies at point X in the Brillouin zone, obtained using different numerical schemes. Columns A–C correspond to FEM results using 25, 28, and 30 FG elements, respectively, demonstrating clear mesh convergence as the frequency variation between the 28- and 30-element models remains below 0.03%. Columns D and E report the COMSOL results generated using fine mapped meshes with 4,104 degrees of freedom (Dofs) and 22,641 degrees of freedom, respectively, both subjected to Floquet–Bloch periodic boundary conditions. The use of two mesh densities helps establish mesh independence in the COMSOL framework as well. Column F contains the DSM predictions, which serve as the reference solution for error evaluation. To quantitatively assess the agreement among the methods, the relative error in eigenfrequencies is computed with respect to the DSM results, and the corresponding distribution for the first ten modes is shown in Fig. 10. The FEM errors systematically decrease with mesh refinement, while both COMSOL meshes exhibit deviations within 3%, affirming the consistency between converged FEM, COMSOL, and DSM solutions. These results collectively confirm the reliability of the numerical implementation and the accuracy of the DSM formulation for FG periodic lattices. Additionally, Table 3 presents a comparison of the 1st, 5th, and 10th eigenfrequencies of an FG diamond lattice structure, computed using the DSM, FEM, and COMSOL Multiphysics. The comparison is conducted at the principal high-symmetry points Γ , and M in the Brillouin zone.

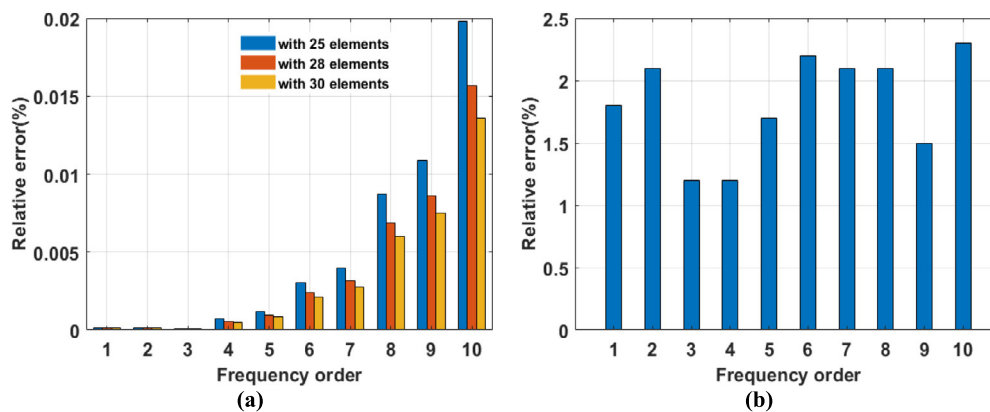
4.2 FG diamond lattice structure

Figure 11a and b depicts the dispersion curves of the FG-DL for the first five modes at power-law indices $p = 0$ and $p = 5$, respectively. The dispersion curves computed using the DSM are shown in blue, while those obtained via FEM are shown in red for validation purposes. A significant reduction in the frequency magnitudes is observed across all modes with increasing p , although the overall shape of the dispersion curves remains unchanged. This frequency shift is attributed to the increased metallic content in the beam elements

Table 2 Comparison of eigenfrequencies at point X ($\beta_1 = 0, \beta_2 = \pi$) in the Brillouin zone, as shown in Fig. 9

Frequency order	A FEM (25 elements per FG beam)	B FEM (28 elements per FG beam)	C FEM (30 elements per FG beam)	D COMSOL (CM* with 4104 Dofs)	E COMSOL (FM* with 22,641 Dofs)	F DSM (1 element per FG beam)
1	534.408743	534.408736	534.408733	545.36234	544.02730	534.407958
2	1454.245676	1454.245468	1454.245364	1489.7256	1484.78279	1454.243673
3	2239.099542	2239.098821	2239.098459	2224.0499	2212.22778	2239.096949
4	4691.112089	4691.105124	4691.101635	4654.5806	4634.78552	4691.078468
5	6434.614398	6434.598244	6434.590147	6580.2241	6543.92496	6434.537822
6	9722.156247	9722.094416	9722.063440	10,013.9937	9935.73984	9721.858951
7	11,232.423095	11,232.329973	11,232.283313	11,077.7513	10,996.1052	11,231.976765
8	16,481.431257	16,481.131125	16,480.980737	16,286.9771	16,133.9147	16,479.994607
9	18,944.827588	18,944.398006	18,944.182680	19,403.0622	19,226.90532	18,942.763863
10	24,886.498110	24,885.469496	24,884.953980	25,700.2123	25,453.8445	24,881.568487

FM* → Fine Meshing, CM* → Coarse Meshing

**Fig. 10** Bar graph showing the relative errors in FEM results with respect to DSM results at various discretization levels, based on the data from Table 2 at high-symmetry point X **b** Bar graph showing the relative errors in COMSOL results with respect to DSM results, based on the data from Table 2 at high-symmetry point X**Table 3** Eigenfrequencies (in rad/s) evaluated at the high-symmetry points of the Brillouin zone, as shown in Fig. 9

High-symmetry points	Frequency order	FEM (30 elements per FG beam)	COMSOL	DSM
Γ - point ($\beta_1 = 0, \beta_2 = 0$)	1	630.259087	641.604689	630.260009
	5	3734.699888	3798.179270	3734.689547
	10	18,377.599518	18,798.883955	18,376.230650
M—point ($\beta_1 = \pi, \beta_2 = \pi$)	1	932.293251	949.074987	932.293701
	5	8316.202292	8457.447238	8316.073980
	10	27,136.383781	27,756.032635	27,131.996711

 β_1 and β_2 are the wavevectors

at higher power-law indices, which results in a decrease in both stiffness and density of the lattice material, thereby lowering the natural frequencies.

Moreover, an evident reduction in bandgap size is observed as the index of material gradient increases. Specifically, the bandgap of the FG-DL structure decreases by approximately 36%, from 700 (rad/s) to 446 (rad/s), as p increases from 0 to 5. This trend is primarily due to the more pronounced frequency reduction in higher modes compared to lower modes. Figure 11c and d further supports this observation by illustrating the effect of the index of material gradient on the 3rd and 8th modes, respectively. The 8th mode exhibits a more substantial frequency drop than the 3rd mode, indicating that higher-order modes are more sensitive to material gradation.

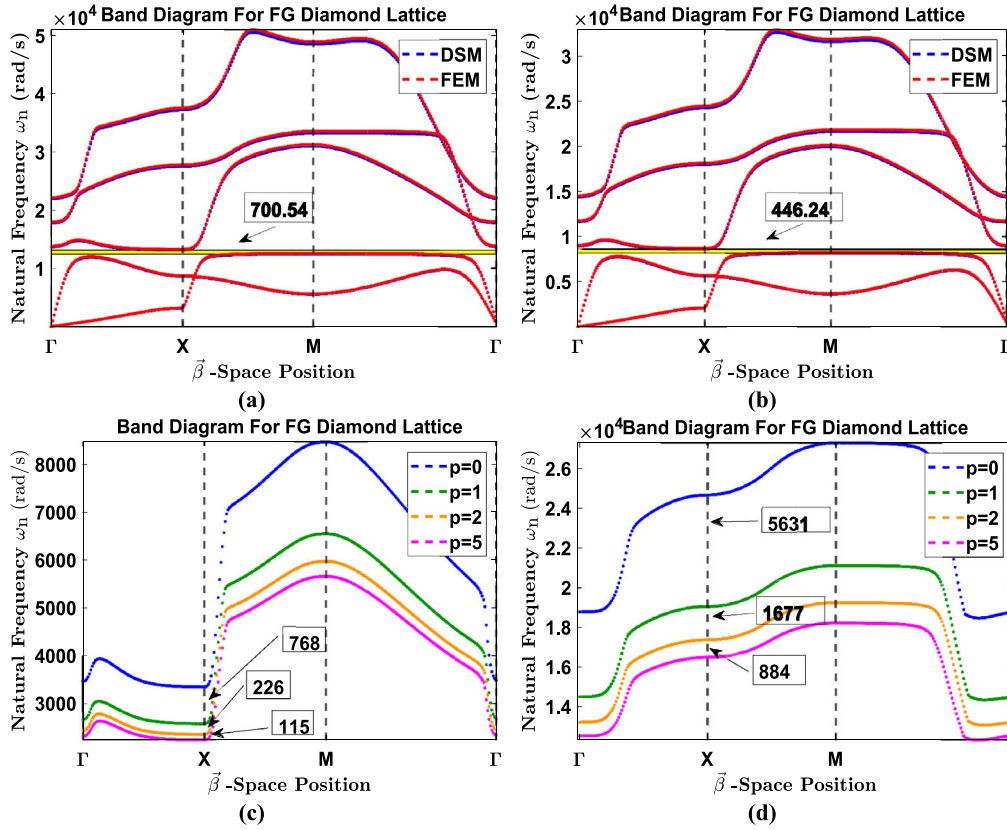


Fig. 11 FG-DL dispersion plots at $h = L/5$ for power-law indices: **a** $p = 0$, **b** $p = 5$ and impact of variation in index of material gradient on **c** the 3rd mode **d** the 8th mode dispersion curve of the FG-DL with $h = L/10$

Figure 12a–c presents the iso-frequency contours of mode 1 for the functionally graded diamond lattice (FG-DL) structure under varying values of the index of material gradient p . Iso-frequency contours—defined as curves of constant frequency in β -space (wavevector space)—represent cross-sectional “slices” of the dispersion relation at specific frequencies, and offer insight into wave-propagation directionality. As shown in Fig. 12a–c increasing the index of material gradient from $p = 0$ to $p = 1$ leads to a noticeable increase in the density and complexity of the iso-frequency contours. This suggests enhanced anisotropy and more pronounced directional wave propagation. Interestingly, beyond $p = 1$, the contour density begins to decrease, indicating a shift in wave-propagation characteristics due to the altered balance of stiffness and mass in the graded material. Additionally, the transition from circular to non-circular contour shapes with increasing p further reflects the emergence of directionally dependent wave behavior in the FG-DL structure.

Figure 12d–f illustrates the influence of the index of material gradient on the phase velocity of the FG-DL. As the exponent of the material gradient increases, a consistent reduction in phase velocity is observed across all modes. This trend is attributed to the corresponding decrease in both stiffness and density of the FG-DL, which lowers the wave-propagation speed. The ability to modulate phase velocity through variation of the gradient index p highlights the tunability of FG-DL structures. These features make them attractive for adaptive metamaterial applications. Additionally, for modes 1, 3, and 7, the phase velocity exhibits less pronounced reductions along certain propagation directions, indicating directional sensitivity. This anisotropic behavior can be utilized in the design of waveguides or phononic devices that require directional control of wave propagation.

Figure 13 presents the directional percentage reduction in phase speed of FG-DL structure when the index of material gradient is increased from $p = 0$ to $p = 1$. For each propagation direction θ , the phase-speed

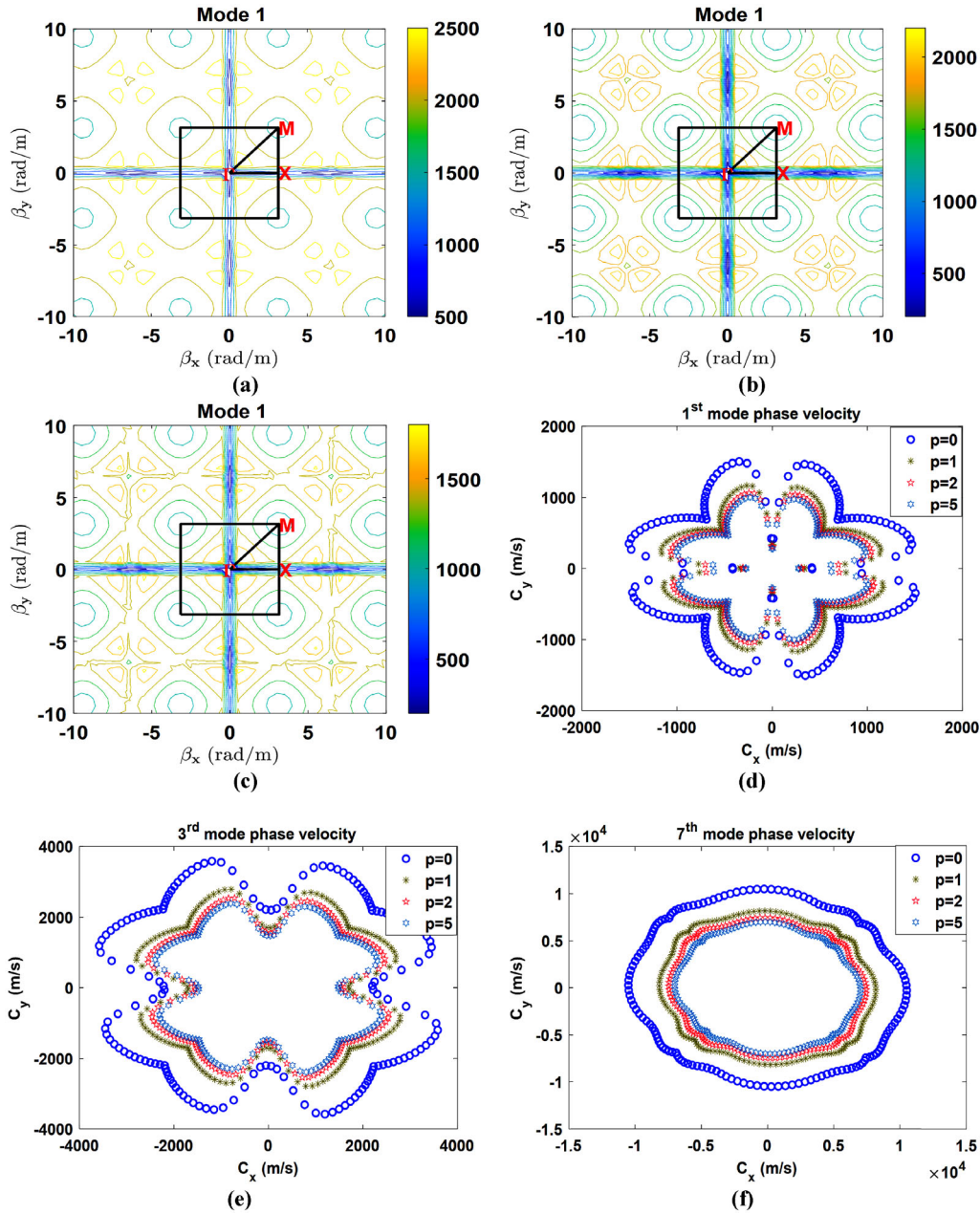


Fig. 12 The first mode FG-DL structure iso-frequency contours with **a** $p = 0$ **b** $p = 1$ **c** $p = 10$ and variations in phase velocity for the FG-DL in the following modes: **e** Mode 1, **f** Mode 3, and **g** Mode 7 (* C_x and C_y denote phase velocities along the x - and y -directions, respectively)

magnitude is obtained from the vector components C_x and C_y , i.e., $C(\theta) = \sqrt{C_x^2 + C_y^2}$. The reduction ratio is then evaluated using reduction formula

$$R(\theta) = \frac{C_0(\theta) - C_1(\theta)}{C_0(\theta)} \times 100\% \quad (34)$$

The resulting polar plot clearly indicates non-uniform attenuation of the phase speed across directions, reflecting the intrinsic anisotropy introduced by the graded material distribution.

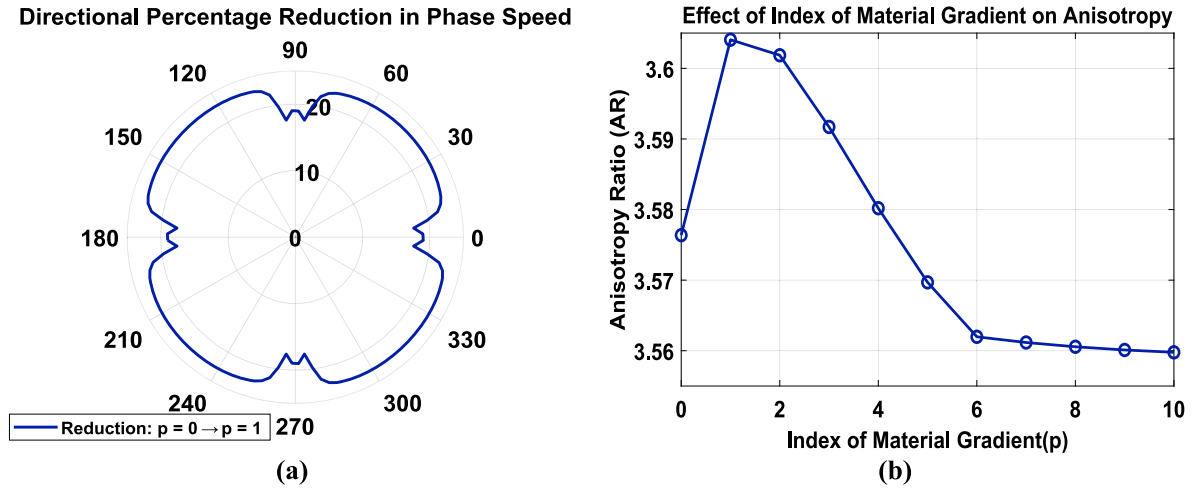


Fig. 13 Directional wave-propagation characteristics of the FG-DL structure. **a** Directional percentage reduction in phase speed between $p = 0$ and $p = 1$, **b** anisotropy ratio of the phase-speed distribution

The anisotropic behavior of the FG-DL is quantified directly from the iso-frequency contour by evaluating the directional variation of the wavevector magnitude. For a fixed frequency ω the anisotropy ratio (AR) is computed as is evaluated as

$$AR = \frac{\beta_{\max}}{\beta(\theta)} \quad (35)$$

where $\beta(\theta)$ denotes the radius of the iso-frequency contour (i.e., the wavevector magnitude) measured along direction θ , and $\beta_{\max}$ is the maximum wavevector value recorded over all directions. An AR value of unity corresponds to completely isotropic wave propagation, while larger values indicate increasing directional dependence of the phase speed. The variation of AR with the index of material gradient p , shown in Fig. 13, reveals that anisotropy initially increases for smaller p , reaches a peak around $p = 1 - 2$, and subsequently decreases and stabilizes at higher gradient indices. This behavior reflects how material gradation modifies the stiffness distribution and thus the directional wave-propagation characteristics.

The dispersion characteristics for the FG-DL under the exponential and trigonometric gradation profiles are presented in Fig. 14. Subplots (a) and (c) correspond to the exponential profile for $p = 1$ and $p = 5$, respectively, while subplots (b) and (d) show the trigonometric profile for the same gradation indices. Examination of the first ten modes indicates that the trigonometric profile produces slightly higher modal frequencies compared to the exponential profile. This behavior arises from the nature of the trigonometric gradation, which results in a comparatively higher effective stiffness for identical values of p . The enhanced stiffness shifts the dispersion branches upward and also leads to a wider bandgap region relative to the exponential model.

Across both gradation profiles, increasing the index of material gradient from $p = 1$ to $p = 5$ results in a clear downward shift of the dispersion branches. This monotonic reduction is attributed to the increasing metallic content at higher p -values, which lowers the effective stiffness and, consequently, reduces wave speeds. From a design standpoint, these observations highlight the sensitivity of wave propagation to the chosen gradation law: The trigonometric profile provides higher cut-off frequencies and broader band gaps desirable for high-stiffness or vibration-filtering applications, while the exponential profile yields comparatively softer, more attenuative responses. This comparative assessment provides a useful basis for tailoring graded structures for specific dynamic performance requirements.

4.3 FG chiral lattice structure

Band diagram Calculation of FG chiral lattice is considered next. Here, the unit cell of chiral lattice consists of 8 beam elements, each with a length L . The cross section of each beam is square with a side length of $h = L/15$. The material characteristics of the beam are regarded as identical to those of the previous one. The dispersion curves of the FG chiral lattice for the first 10 modes at power law indices $p = 5$ and $p = 10$ are shown in

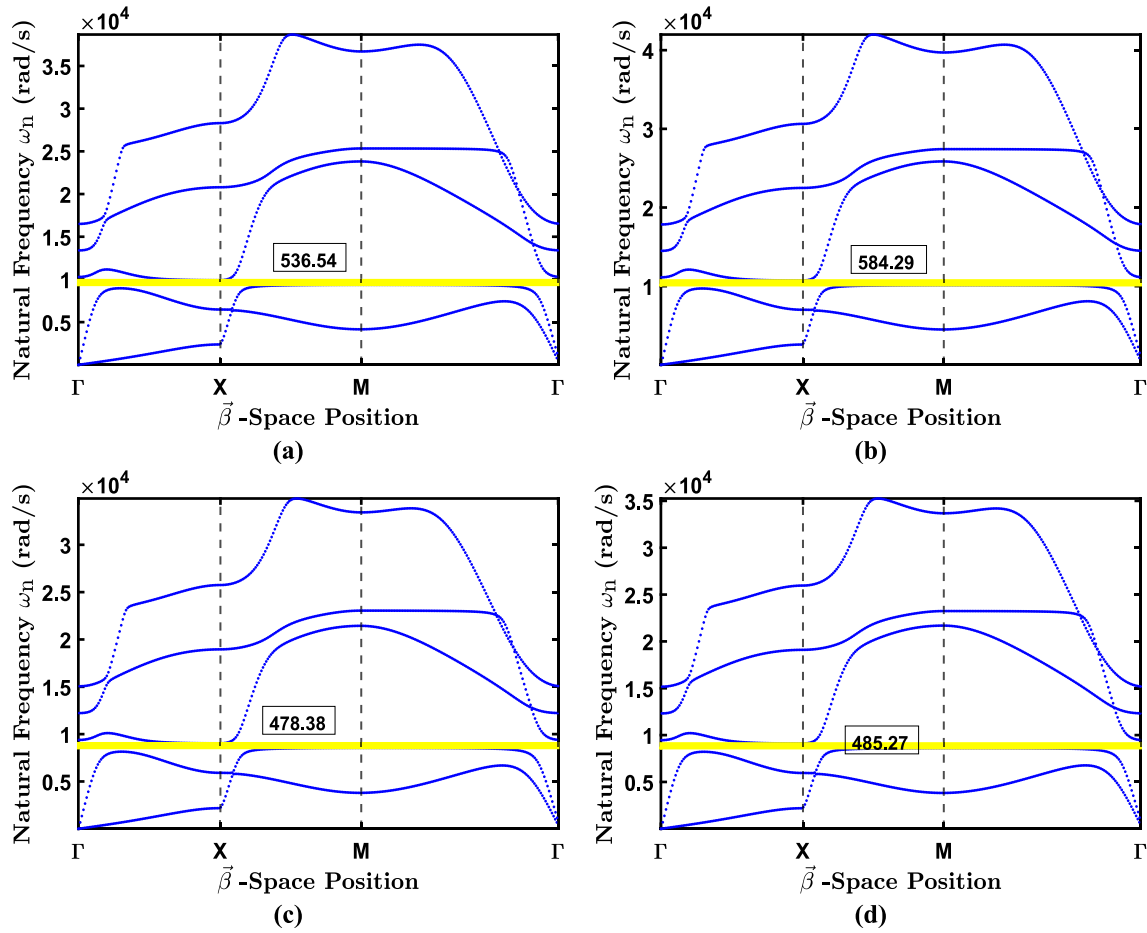


Fig. 14 Dispersion curves of FG-DL for different gradation laws: **a** exponential law with $p = 1$, **b** trigonometric law with $p = 1$, **c** exponential law with $p = 5$, and **d** trigonometric law with $p = 5$

Fig. 15. The finite element method (shown in red) and the proposed DSM approach (shown in blue) are used to create the curves to simulate the chiral lattice in the FEM framework, each individual beam is divided into 30 elements, leading to a cumulative count of 240 elements for the complete unit cell representation.

Figure 15a and b demonstrates the influence of material gradation on the dynamic response of the FG chiral lattice. Figure 15a and b shows that as the index of material gradient increases, the bandgaps progressively narrow, reflecting a downward shift in the dispersion characteristics. In the FG chiral lattice structure, all four band gaps show a consistent reduction in size, collectively decreasing by approximately 32% as the index of material gradient increases from $p = 0$ to $p = 5$. This reduction is slightly less pronounced compared to the 36% decrease observed in the FG-DL structure. Figure 15c and d further clarifies this behavior by isolating specific mode frequencies (3rd and 8th), where the higher modes show a more significant decrease.

Figure 16a–c illustrates the constant frequency curves for the first mode of the FG chiral lattice, showing how they change with different values of the index of material gradient (p). For the FG chiral lattice, the iso-frequency contour density—indicated by the number of circular or non-circular contour curves—decreases as the index of material gradient increases from $p = 0$ to $p = 1$, and then rises again at $p = 10$. This non-monotonic trend differs from the FG diamond lattice, where the contour density initially increases from $p = 0$ to $p = 1$ and subsequently decreases as the index increases to $p = 10$.

Figure 16d–f demonstrates the impact of the index of material gradient on the phase velocity of the FG chiral lattice structure. As observed in the figure, an increase in the index of material gradient leads to a consistent reduction in the phase velocity across all modes. This trend is explained by the changes in the material properties that occur with varying the exponent of material gradient. Specifically, as the index of material gradient rises, the FG chiral lattice exhibits a decrease in both stiffness and density. Consequently,

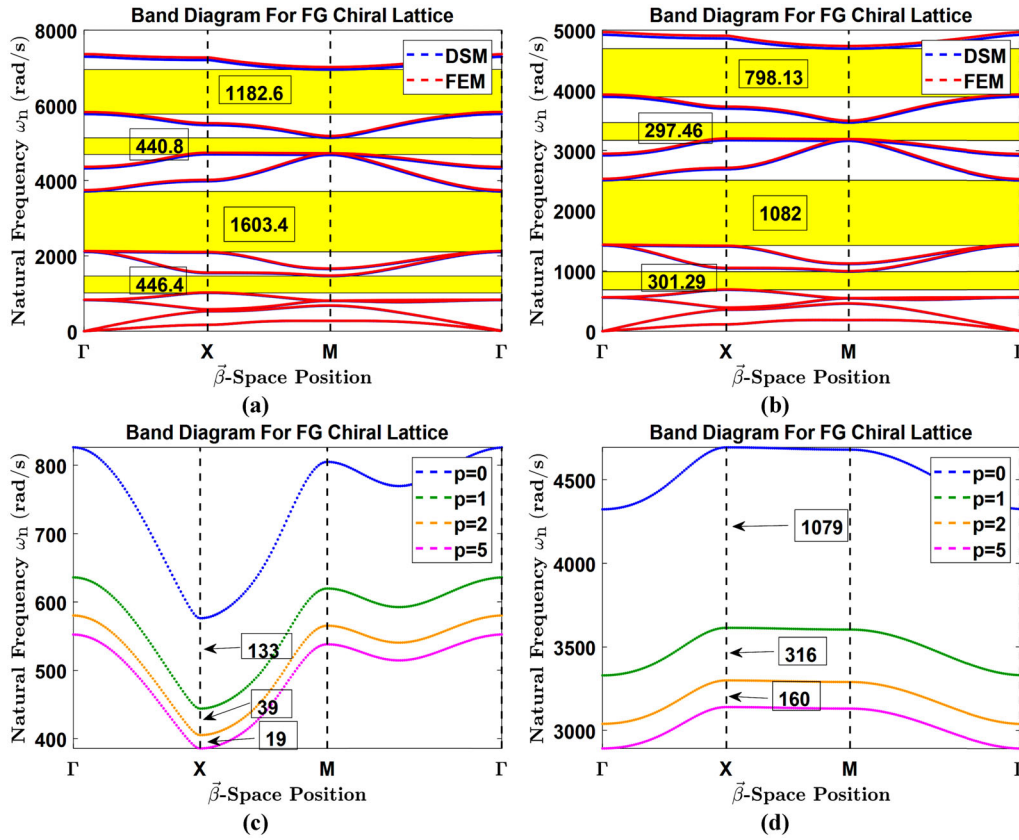


Fig. 15 Dispersion curves of FG chiral lattice with **a** $p = 0$ **b** $p = 5$ and effect of variation in index of material gradient on **c** the 3rd mode **d** the 8th mode dispersion curves of the FG chiral lattice with $h = L/15$ ($L = 0.125$ m)

these changes in material properties collectively contribute to a lower phase velocity, reflecting the slower wave propagation within the lattice as the index of material gradient increases.

Figure 17a illustrates the directional percentage reduction in phase speed for the FG chiral lattice when the index of material gradient is increased from $p = 0$ to $p = 1$. Compared with the behavior reported earlier for the FG-DL configuration in Sect. 4.2, the FG chiral structure exhibits a smoother yet directionally biased reduction pattern, reflecting the influence of its inherent geometric chirality. This response indicates that the graded chiral architecture modifies wave-propagation behavior in a non-uniform manner across different directions, highlighting the strong interplay between material gradation and microstructural symmetry.

For the FG chiral lattice, the anisotropy ratio is evaluated from the directional variation in phase speed over the full propagation domain. As shown in Fig. 17b, a slight reduction is observed up to $p = 1$, after which the ratio increases steadily with the index of material gradient. This trend contrasts with the behavior reported for the diamond lattice in Sect. 4.2, where the anisotropy ratio begins to decrease beyond $p = 1$. The chiral configuration therefore exhibits a progressively strengthening directional response with increasing gradation, while simultaneously maintaining relatively small overall variations, indicating that its dynamic anisotropy is less sensitive to grading compared to the other lattices.

Figure 18 illustrates the dispersion behavior of the chiral lattice for the exponential and trigonometric gradation profiles at $p = 1$ and $p = 5$. Across all subplots, the trigonometric profile consistently produces marginally higher modal frequencies than the exponential profile, owing to its comparatively stiffer effective gradation. This increased stiffness also contributes to a wider bandgap region in the trigonometric case. For both gradation schemes, increasing the index of material gradient from $p = 1$ to $p = 5$ leads to a systematic reduction in the dispersion branches, a result of the enhanced metallic enrichment that lowers the overall rigidity. These trends indicate that, despite the presence of chiral geometry, the material gradation law remains the dominant factor governing the frequency spectrum within the examined mode rang.

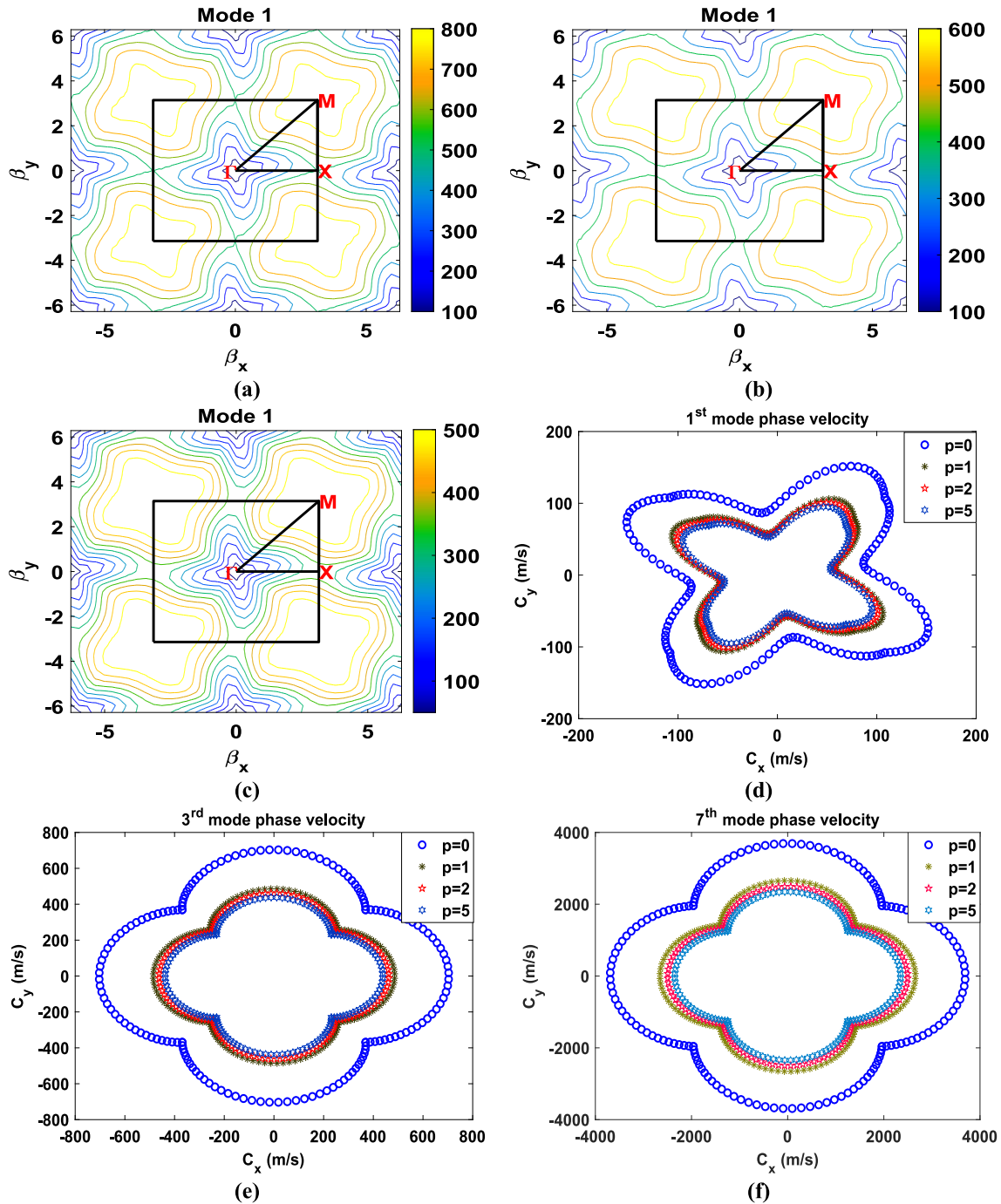


Fig. 16 The first mode FG chiral lattice structure iso-frequency contours with **a** $p = 0$ **b** $p = 1$ **c** $p = 10$ and variations in phase velocity for the FG chiral lattice in the following modes: **d** Mode 1, **e** Mode 3, and **f** Mode 7

4.4 FG triangular lattice structure

The next analysis step involves calculating the dispersion curve for the FG triangular lattice. The unit cell in this case consists of three beam elements, each with a specified length L . The cross section of each beam is square-shaped, with a side length given by $h = L/15$. The material properties of the beams are consistent with those used in earlier sections of this study. Figure 19 presents the computed dispersion curves for the FG triangular lattice, focusing on the first ten modes at power law indices $p = 0, 1, 2$, and 5 . The band structures

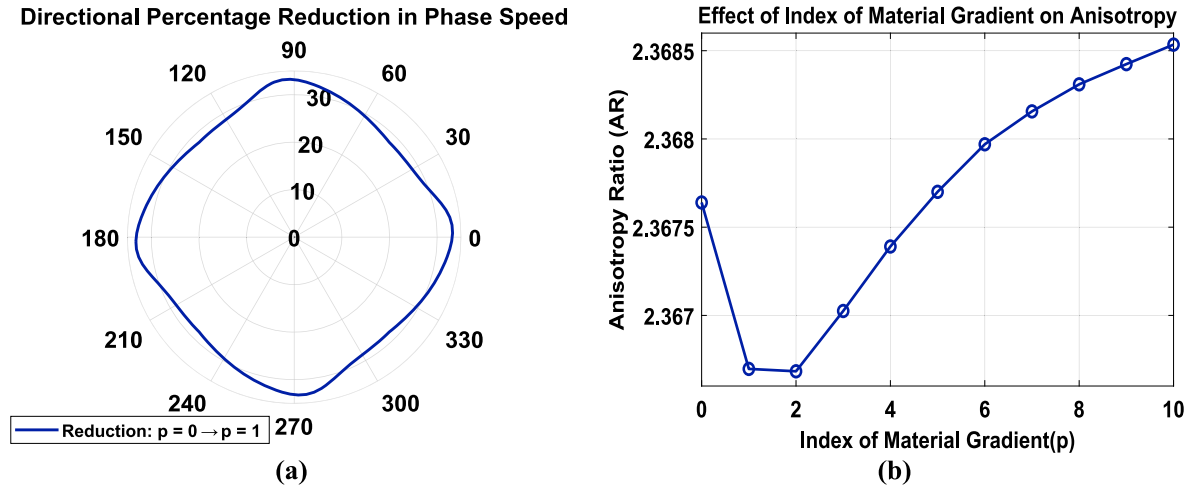


Fig. 17 Directional wave-propagation characteristics of the FG chiral lattice. **a** Directional percentage reduction in phase speed between $p = 0$ and $p = 1$, **b** anisotropy ratio of the phase-speed distribution

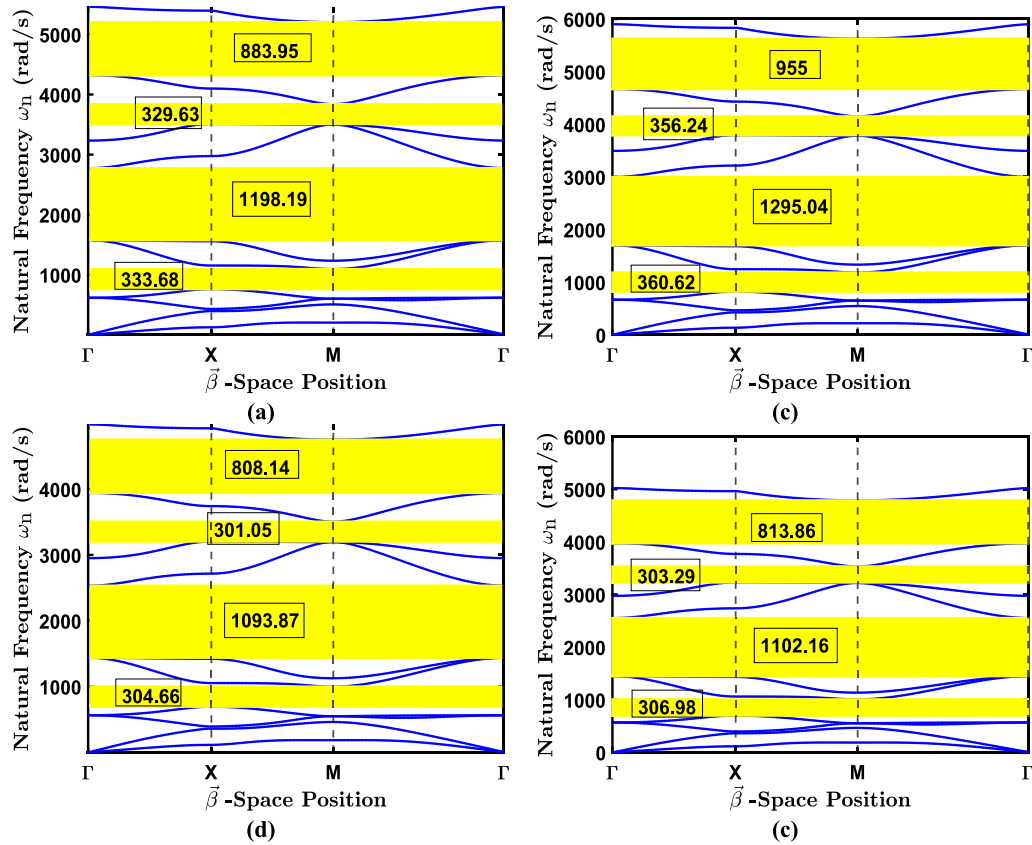


Fig. 18 Dispersion curves of FG chiral lattice for different gradation laws: **a** Exponential law with $p = 1$, **b** trigonometric law with $p = 1$, **c** exponential law with $p = 5$, and **d** trigonometric law with $p = 5$

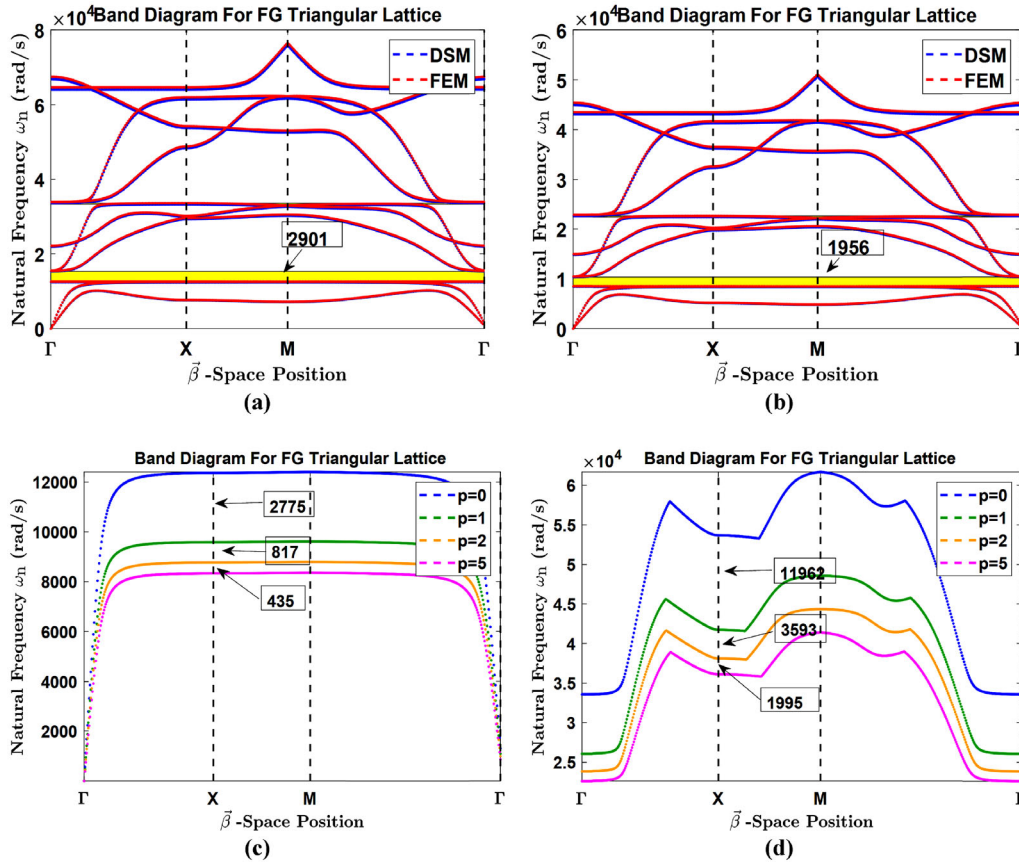


Fig. 19 Band diagram of FG triangular lattice **a** $p = 0$ **d** $p = 5$ and effect of variation in index of material gradient on **c** the 3rd mode **d** the 8th mode dispersion curves of the FG triangular lattice with $h = L/15$ ($L = 0.125$ m)

have been determined using the DSM, shown in blue and cross-verified using the FEM, shown in red. The unit cell of the triangular lattice is discretized into 30 beam elements per beam for the FEM analysis, yielding 90 beam elements for the unit cell as a whole.

Figure 19a and b compares the dispersion curves of the FG triangular lattice for two power-law indices $p = 0$ and 5. The figure illustrates that as the index of material gradient increases; both the frequency magnitudes and the band gap sizes decrease. Specifically, the band gap size reduces by approximately 32%, consistent with the FG chiral lattice but slightly less than the 36% reduction observed in the FG diamond lattice. This trend highlights the tunable nature of the FG triangular structure with respect to material gradation.

The band structures for the FG triangular lattice's third, fifth, seventh, and eighth modes at different power-law indices are shown in Fig. 19c and d. The figure reveals that the reduction in frequency magnitude becomes more pronounced in the higher modes compared to the lower ones, indicating that higher-order wave modes are more sensitive to variation in material gradation.

Figure 20 depicts the impact of the index of material gradient on the phase velocity of the FG triangular lattice. The FG triangular lattice shows a consistent trend across all modes, which is comparable to the findings in FG chiral and diamond lattices: the phase velocity reduces as the index of material gradient increases.

Figure 21a depicts the directional percentage reduction in phase speed for the FG triangular lattice when the gradation index is increased from $p = 0$ to $p = 1$. In contrast to the FG-DL and FG chiral configurations discussed earlier, the triangular architecture displays a more uniform reduction pattern, with only mild angular variation across the propagation directions. This behavior reflects the higher geometric symmetry of the triangular unit cell, which limits directional bias in the graded response. Overall, the results indicate that the introduction of material gradation affects the triangular lattice in a comparatively isotropic manner, leading to a consistent decrease in phase speed over the full angular range.

For the FG triangular lattice, the anisotropy ratio follows a trend similar to that of the diamond lattice. As shown in Fig. 21b, the anisotropy increases up to $p = 1$ and then decreases steadily as the material gradient

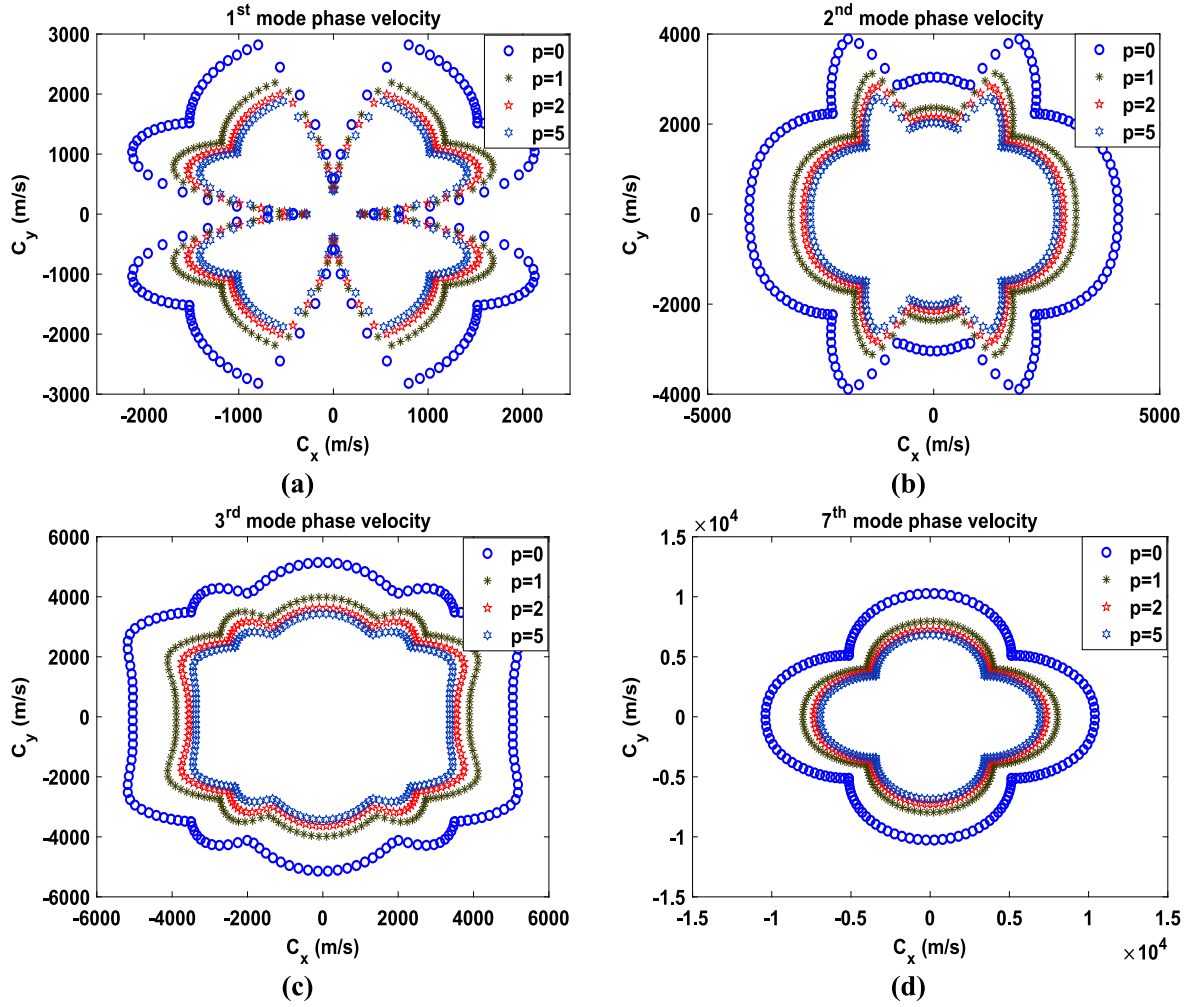


Fig. 20 Variations in phase velocity for the FG triangular lattice in the following modes: **a** Mode 1, **b** Mode 2, **c** Mode 3, and **d** Mode 7

index increases. However, the overall magnitude remains different from the diamond case, reflecting the distinct geometric influence of the triangular layout. The anisotropy ratio is obtained from the iso-frequency contour by evaluating the directional variation of the wavevector magnitude. This behavior indicates that, although the qualitative trend matches that of the diamond lattice, the triangular configuration exhibits a comparatively lower directional contrast across the examined range of p .

Figure 22 shows the dispersion response of the triangular lattice under the exponential and trigonometric gradation schemes for $p = 1$ and $p = 5$. Across all subplots, the trigonometric profile yields slightly higher modal frequencies than the exponential profile, reflecting its comparatively stiffer effective gradation. The trigonometric scheme also exhibits a broader bandgap region. For both gradation laws, increasing the index of material gradient from $p = 1$ to $p = 5$ produces a consistent downward shift in the dispersion curves due to the greater metallic enrichment at higher p -values, which reduces the overall stiffness. The triangular lattice therefore demonstrates the same qualitative sensitivity to material gradation observed in the other topologies, confirming that the dispersion behavior is governed more strongly by the gradation law than by the lattice geometry within the examined frequency range.

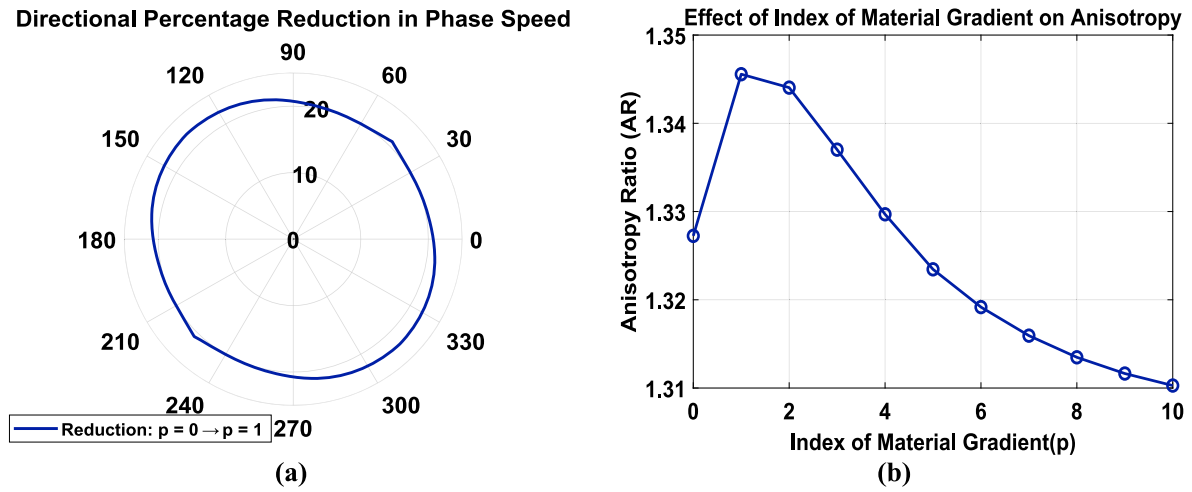


Fig. 21 Directional wave-propagation characteristics of the FG triangular lattice. **a** Directional percentage reduction in phase speed between $p = 0$ and $p = 1$, **b** anisotropy ratio of the phase-speed distribution

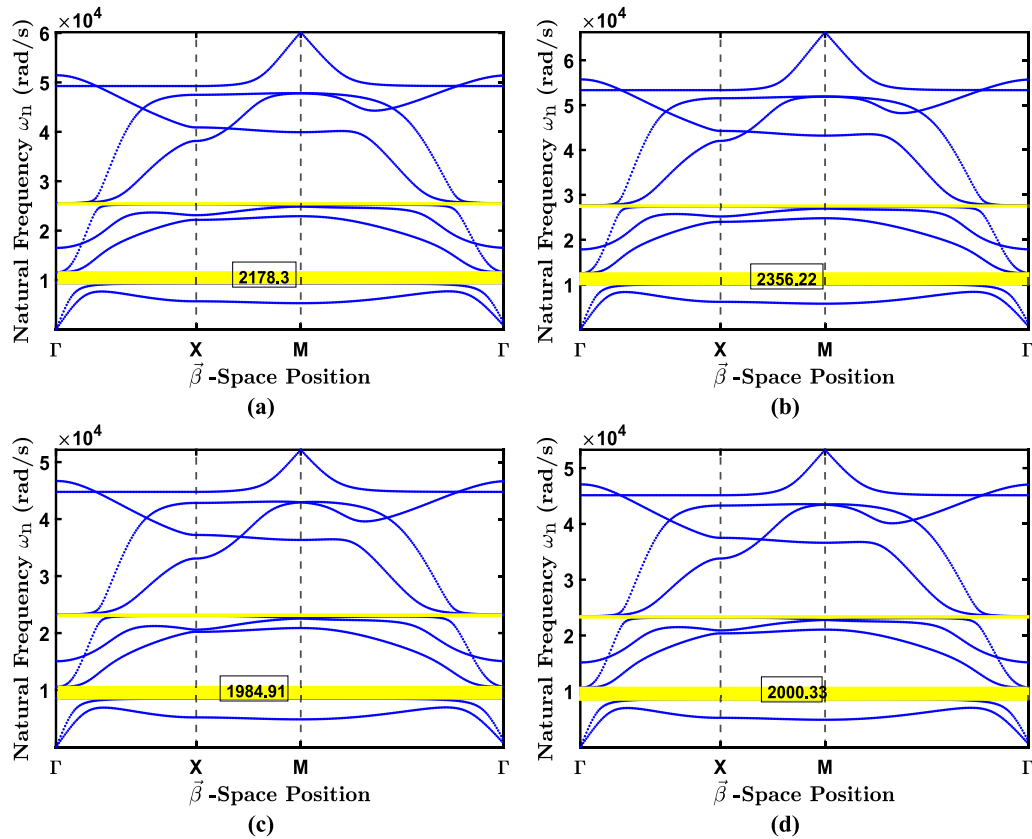


Fig. 22 Dispersion curves of FG triangular lattice for different gradation laws: **a** Exponential law with $p = 1$, **b** trigonometric law with $p = 1$, **c** exponential law with $p = 5$, and **d** trigonometric law with $p = 5$

4.5 Comparison of bandgap behavior across lattice topologies

Among the three lattice configurations, notable differences arise in both the location, width, and directional characteristics of the bandgap regions, each carrying significant engineering implications. The diamond and triangular lattices exhibit band gaps at relatively high frequencies, typically above 10,000 rad/s. Between these two, the triangular lattice demonstrates a substantially wider band gap—approximately 3000 rad/s—compared to the diamond lattice, where the width is only about 700 rad/s, indicating its superior ability to attenuate high-frequency waves over a broader spectral range. In contrast, the chiral lattice develops its primary band gap at much lower frequencies, with the first gap appearing below 1000 rad/s, making it particularly attractive for low-frequency vibration and wave isolation, a capability rarely observed in periodic beam-based lattices.

In terms of anisotropy behavior, the three lattices also show distinct trends. For the diamond and triangular lattices, the anisotropy ratio decreases with increasing gradation index, suggesting a shift toward more isotropic wave behavior as material inhomogeneity increases. Conversely, the chiral lattice exhibits an increasing anisotropy ratio, indicating stronger direction-dependent wave propagation with higher gradient indices. These contrasting trends reveal how lattice topology fundamentally governs the directional dispersion characteristics, even under identical gradation laws.

These outcomes also highlight a clear distinction from the square lattice examined by Mishra and Kumar [40]. In their study, the square lattice exhibits no band gaps within the first ten modes, and a band gap appears only after imposing geometric heterogeneity (different element cross sections), occurring at a very high frequency of nearly 50,000 rad/s. In contrast, the present diamond, triangular, and chiral lattices naturally possess band gaps due solely to their topology, and these gaps occur in significantly lower and more practical frequency ranges. Moreover, the introduction of graded material laws in Eqs. (1) and (2), further shifts these gaps to even lower frequencies, offering an additional tuning mechanism.

Overall, these distinctions demonstrate that lattice geometry, together with material gradation, can be strategically leveraged to target specific operational frequency ranges—from low-frequency isolation using chiral layouts to broad high-frequency filtering using the triangular lattice, with intermediate behavior provided by the diamond configuration.

5 Conclusion

The dynamic behavior of three FG lattice structures—diamond, chiral, and triangular—is investigated using the exact DSM in conjunction with power-law-based material gradation. This research focuses on the influence of varying the index of material gradient on key wave-propagation characteristics such as natural frequencies, band gap size, iso-frequency contours, and phase velocity. The analysis provides insights into how material gradation affects the vibrational and wave-propagation responses of periodic lattice structures. The main conclusions of this study are:

- Increasing the index of material gradient results in a consistent decrease in natural frequencies for all three FG lattice structures.
- Higher-order modes exhibit a more significant reduction in frequency compared to lower-order modes, emphasizing the influence of gradation on dynamic stiffness at elevated modes.
- A reduction in band gap size is observed—approximately 36% for the diamond lattice and around 32% for both the chiral and triangular lattices—as the index of material gradient increases.
- Iso-frequency contour plots revealed more non-circular curves with higher power-law indices, indicating enhanced anisotropy and increased directionality in wave propagation.
- The anisotropy ratio trends across the diamond, chiral, and triangular lattices enable precise control of direction-dependent wave propagation, supporting applications in vibration isolation, wave steering, impact mitigation, and tunable frequency-selective metamaterial design.
- Phase velocity curves showed a decreasing trend with increasing index of material gradient across all lattice types, further confirming the gradation-induced modifications in wave-propagation behavior.
- The directional percentage reduction plot shows that FG-DL, chiral, and triangular lattices exhibit different attenuation characteristics, demonstrating how material gradation can be tailored to achieve direction-dependent control for wave steering and vibration isolation applications.

Although the findings are encouraging, the present study focuses solely on the influence of material property variation along the transverse direction on wave propagation in lattice structures. Future investigations could

extend this work by examining the effects of material gradation along the longitudinal direction to gain a more comprehensive understanding of wave behavior in functionally graded lattices.

In summary, this study highlights the crucial role of functionally graded materials in tuning the wave-propagation characteristics of lattice systems and introduces an effective approach for dispersion analysis applicable to the design of advanced functionally graded lattices.

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Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Appendix A

- Detailed derivation of the governing differential equation for longitudinal vibration of an FG beam

$$\begin{aligned}
 & \delta \left\{ \int_{t_1}^{t_2} \left(\frac{1}{2} \int_0^L M_0 \left(\frac{\partial \hat{u}}{\partial t} \frac{\partial \delta \hat{u}}{\partial t} \right) dx - \frac{1}{2} \int_0^L N_{xx} \delta e_{xx}^{(0)} dx \right) dt \right\} = 0 \\
 & \Rightarrow \left[\int_0^L \left\{ \int_{t_1}^{t_2} M_0 \left(\frac{\partial \hat{u}}{\partial t} \frac{\partial \delta \hat{u}}{\partial t} \right) dt \right\} dx \right] - \left[\int_{t_1}^{t_2} \left\{ \int_0^L N_{xx} \frac{\partial \delta \hat{u}}{\partial x} dx \right\} dt \right] = 0 \\
 & \Rightarrow \left[\int_0^L \left\{ - \int_{t_1}^{t_2} \frac{d}{dt} \left(M_0 \frac{\partial \hat{u}}{\partial t} \right) \delta \hat{u} dt + \rho A_e \frac{\partial \hat{u}}{\partial t} \delta \hat{u} \right\} dx \right] - \left[- \int_{t_1}^{t_2} \left\{ \int_0^L \frac{d}{dx} (N_{xx}) \delta \hat{u} dx + N_{xx} \delta \hat{u} \Big|_0^L \right\} dt \right] = 0 \\
 & \Rightarrow - \int_0^L \int_{t_1}^{t_2} \frac{d}{dt} \left(M_0 \frac{\partial \hat{u}}{\partial t} \right) \delta \hat{u} dt dx + \int_0^L \left(M_0 \frac{\partial \hat{u}}{\partial t} \right) \delta \hat{u} \Big|_{t_1}^{t_2} dx + \int_0^L \int_{t_1}^{t_2} \frac{d}{dx} N_{xx} \delta \hat{u} dt dx - \int_{t_1}^{t_2} \left(N_{xx} \delta \hat{u} \Big|_0^L \right) dt = 0 \\
 & \Rightarrow \left[\int_0^L \int_{t_1}^{t_2} \left\{ - \frac{d}{dt} \left(M_0 \frac{\partial \hat{u}}{\partial t} \right) + \frac{d}{dx} N_{xx} \right\} \delta \hat{u} dt dx \right] + \int_0^L \left(M_0 \frac{\partial \hat{u}}{\partial t} \right) \delta \hat{u} \Big|_{t_1}^{t_2} dx - \left[\int_{t_1}^{t_2} \left(N_{xx} \delta \hat{u} \Big|_0^L \right) dt \right] = 0
 \end{aligned}$$

Governing Differential Equation
Natural Boundary Condition

Appendix B

- Detailed derivation of the governing differential equation for transverse vibration of an FG beam

$$\begin{aligned}
 & \Rightarrow \left\{ \int_{t_1}^{t_2} \left(\int_0^L \left[M_0 \left(\frac{\partial w}{\partial t} \frac{\partial \delta w}{\partial t} \right) + M_2 \frac{\partial \xi}{\partial t} \frac{\partial \delta \xi}{\partial t} \right] dx - \int_0^L \left[M_{xx} \delta e_{xx}^{(1)} + Q_x \delta \gamma_{xz} \right] dx \right) dt \right\} = 0 \\
 & \Rightarrow \left\{ \int_{t_1}^{t_2} \left(\int_0^L \left[M_0 \left(\frac{\partial w}{\partial t} \frac{\partial \delta w}{\partial t} \right) + M_2 \frac{\partial \xi}{\partial t} \frac{\partial \delta \xi}{\partial t} \right] dx - \int_0^L \left[M_{xx} \frac{\partial \delta \xi}{\partial x} + Q_x \left(\frac{\partial \delta w}{\partial x} + \delta \xi \right) \right] dx \right) dt \right\} = 0 \\
 & \Rightarrow \left[\int_0^L \left\{ - \int_{t_1}^{t_2} \frac{d}{dt} \left(M_0 \frac{\partial w}{\partial t} \delta w + M_2 \frac{\partial \xi}{\partial t} \delta \xi \right) dt + \left(M_0 \frac{\partial w}{\partial t} \delta w \right) \Big|_{t_1}^{t_2} + M_2 \frac{\partial \xi}{\partial t} \delta \xi \Big|_{t_1}^{t_2} \right\} dx \right. \\
 & \quad \left. - \left[- \int_{t_1}^{t_2} \left\{ \int_0^L \frac{d}{dx} (M_{xx}) \delta \xi dx + \int_0^L \frac{d}{dx} (Q_x) \delta w dx + \int_0^L (Q_x) \delta \xi dx + \left(M_{xx} \delta \xi \Big|_0^L + Q_x \delta w \Big|_0^L \right) \right\} dt \right] \right] = 0 \\
 & - \int_0^L \frac{d}{dt} \left(M_0 \frac{\partial w}{\partial t} \right) \delta w dt dx - \int_0^L \frac{d}{dt} \left(M_2 \frac{\partial \xi}{\partial t} \right) \delta \xi dt dx + \int_0^L \left(\left(M_0 \frac{\partial w}{\partial t} \right) \delta w \Big|_{t_1}^{t_2} \right) dx + \int_0^L \left(\left(M_2 \frac{\partial \xi}{\partial t} \right) \delta \xi \Big|_{t_1}^{t_2} \right) dx \\
 & \quad + \int_0^L \int_{t_1}^{t_2} \frac{d}{dx} (M_{xx}) \delta \xi dt dx + \int_0^L \int_{t_1}^{t_2} \frac{d}{dx} (Q_x) \delta w dt dx + \int_0^L \int_{t_1}^{t_2} Q_x \delta \xi dt dx = 0 \\
 & \Rightarrow - \int_0^L \int_{t_1}^{t_2} \left(\frac{d}{dt} \left(M_0 \frac{\partial w}{\partial t} \right) - \frac{d}{dx} (Q_x) \right) \delta w dt dx - \int_0^L \int_{t_1}^{t_2} \left(\frac{d}{dt} \left(M_2 \frac{\partial \xi}{\partial t} \right) - \frac{d}{dx} (M_{xx}) - Q_x \right) \delta \xi dt dx \\
 & \quad \left. + \int_0^L \left(\left(M_0 \frac{\partial w}{\partial t} \right) \delta w \Big|_{t_1}^{t_2} \right) dx + \int_0^L \left(\left(M_2 \frac{\partial \xi}{\partial t} \right) \delta \xi \Big|_{t_1}^{t_2} \right) dx - \int_{t_1}^{t_2} \left(\left[M_{xx} \delta \xi \right]_0^L + \left[Q_x \delta w \right]_0^L \right) dt = 0
 \end{aligned}$$

Governing Differential Equation
Governing Differential Equation

Natural Boundary Conditions

Appendix C

- Computed parameters for axial vibration of the structure

$$\begin{aligned}
 a_1 &= E_0 \bar{\lambda}_a \cot(\bar{\lambda}_a L) \\
 a_2 &= -E_0 \bar{\lambda}_a \operatorname{cosec}(\bar{\lambda}_a L) \\
 M_0 &= \int_A \rho(z) dA = \frac{bh(\rho_c + p\rho_m)}{1+p} \\
 E_0 &= \int_A E(z) dA = \frac{bh(E_c + pE_m)}{1+p}
 \end{aligned}$$

- Computed parameters for bending vibration of the structure

$$\begin{aligned}
 b_{11} &= R_3 \bar{b}^2 (\lambda_2 + \eta \lambda_1) (cS + \eta sC) / (\lambda_1 \lambda_2 \delta) \\
 b_{12} &= R_2 k_1 [(\lambda_1 + \eta \lambda_2) sS - (\lambda_2 - \eta \lambda_1) (1 - cC)] / \delta \\
 b_{22} &= R_1 (\lambda_2 + \eta \lambda_1) (sC - \eta cS) / \delta \\
 b_{13} &= -R_3 \bar{b}^2 (\lambda_2 + \eta \lambda_1) (S + \eta s) / (\lambda_1 \lambda_2 \delta) \\
 b_{14} &= R_2 k_1 (\lambda_2 + \eta \lambda_1) (C - c) / \delta \\
 b_{24} &= R_1 (\lambda_2 + \eta \lambda_1) (\eta S - s) / \delta
 \end{aligned}$$

with

$$\begin{aligned} W_j &= E_e/L^j, \quad j = 1, 2, 3 \\ s &= \sin \lambda_1, \quad c = \cos \lambda_1, \quad S = \sinh \lambda_2, \quad C = \cosh \lambda_2 \\ \eta &= k_1/k_2, \quad \delta = 2\eta(1 - cC) + (1 - \eta^2)sS \end{aligned}$$

where

$$\left. \begin{matrix} \lambda_1 \\ \lambda_2 \end{matrix} \right\} = \bar{b} \left\{ \pm \Delta/2 + \left[\Delta^2/4 + \left(1 - \bar{b}^2 r^2 s^2 \right) / \bar{b}^2 \right]^{1/2} \right\}^{1/2}$$

with

$$\Delta = r^2 + s^2, \quad \bar{b}^2 = \frac{M_2 \omega^2 L^4}{E_0}, \quad r^2 = \frac{I_e}{AL^2}, \quad s^2 = \frac{E_2}{K_s G_0 L^2}$$

where

$$\begin{aligned} I_e &= \int_{A_t} (z - \check{n})^2 d\hat{A} = \frac{bh^3}{12} \left[E_m + 12(E_c - E_m) \left(\frac{1}{p+3} - \frac{1}{p+2} + \frac{1}{4(p+1)} \right) - \frac{3p^2(E_c - E_m)^2}{(p+1)(p+2)^2(E_c + pE_m)} \right] \\ E_2 &= \int_A E(z)(z - \check{n})^2 d\hat{A} = \frac{bh^3}{12} \left\{ E_m \left[4 - 12 \left(\frac{z_o}{h} + \frac{1}{2} \right) + 12 \left(\frac{z_o}{h} + \frac{1}{2} \right)^2 \right] \right. \\ &\quad \left. + (E_c - E_m) \left[\frac{12}{p+3} - \frac{24}{p+2} \left(\frac{z_o}{h} + \frac{1}{2} \right) + \frac{12}{(p+1)} \left(\frac{z_o}{h} + \frac{1}{2} \right)^2 \right] \right\} \\ M_2 &= \int_A \rho(z)(z - \check{n})^2 dA = \left\{ \rho_m \left[4 - 12 \left(\frac{z_o}{h} + \frac{1}{2} \right) + 12 \left(\frac{z_o}{h} + \frac{1}{2} \right)^2 \right] \right. \\ &\quad \left. + (\rho_c - \rho_m) \left[\frac{12}{p+3} - \frac{24}{p+2} \left(\frac{z_o}{h} + \frac{1}{2} \right) + \frac{12}{(p+1)} \left(\frac{z_o}{h} + \frac{1}{2} \right)^2 \right] \right\} \\ \check{n} &= \frac{\int_{-h/2}^{h/2} z E(z) b dz}{E(z) b dz} = \frac{p(E_c - E_m)h}{2(p+2)(E_c + pE_m)} \end{aligned}$$

References

1. Zhang, X., Zhao, Z., Duan, S., Lei, H., Fang, D.: The tensile strength of brittle diamond lattice structure with material dispersion. *J. Appl. Mech.* **91**, 66 (2024)
2. Leamy, M.J.: Exact wave-based Bloch analysis procedure for investigating wave propagation in two-dimensional periodic lattices. *J. Sound Vib.* **331**, 1580–1596 (2012)
3. Cheng, S., Yang, H., Li, X., Ding, Q., Yan, Q., Sun, Y., Xin, Y., Wang, L.: Numerical investigation of low band gap and vibration suppression for novel four-ligament chiral composite metamaterial structures. *Acta Mech.* **234**, 2947–2961 (2023)
4. Zhang, Q., Hong, W., Xu, J., Zhang, Y., Ding, S., Wu, W., Xia, R.: Mechanics and wave propagation characterization of chiral S-shaped auxetic metastructure. *Acta Mech. Solida Sin.* **35**, 571–586 (2022)
5. Joedicke, I., Ghavidelnia, N., Felsch, G., Slesarenko, V.: Addressing manufacturing defects in architected materials via anisotropy: minimal viable case. *Acta Mech.* **235**, 2715–2724 (2024)
6. Li, M., Du, S., Li, F., Jing, X.: Vibration characteristics of novel multilayer sandwich beams: modelling, analysis and experimental validations. *Mech. Syst. Signal Process.* **142**, 106799 (2020)
7. Chen, D.H., Ozaki, S.: Analysis of in-plane elastic modulus for a hexagonal honeycomb core: effect of core height and proposed analytical method. *Compos. Struct.* **88**, 17–25 (2009)
8. Ruzzene, M., Scarpa, F.: Directional and band-gap behavior of periodic auxetic lattices. *Phys. Status Solidi (B)* **242**, 665–680 (2005)
9. Sepehri, S., Jafari, H., Mashhadi, M.M., Yazdi, M.R.H., Fakhrabadi, M.M.S.: Study of tunable locally resonant metamaterials: effects of spider-web and snowflake hierarchies. *Int. J. Solids Struct.* **204**, 81–95 (2020)
10. Miranda, E.J.P., Jr., Dos Santos, J.M.C.: Evanescent Bloch waves and complex band structure in magneto-electroelastic phononic crystals. *Mech. Syst. Signal Process.* **112**, 280–304 (2018)
11. Yu, D., Liu, Y., Qiu, J., Wang, G., Zhao, H.: Complete flexural vibration band gaps in membrane-like lattice structures. *Phys. Lett. A* **357**, 154–158 (2006)
12. Ebrahimi, F., Parsi, M.: Utilizing star-shaped auxetic meta beams for piezoelectric vibration energy harvesting. *Acta Mech.* **66**, 1–25 (2025)
13. Zhang, G.Y., Shen, W., Gu, S.T., Gao, X.-L., Xin, Z.-Q.: Band gaps for elastic flexural wave propagation in periodic composite plate structures with star-shaped, transversely isotropic, magneto-electro-elastic inclusions. *Acta Mech.* **232**, 4325–4346 (2021)

14. Mead, D.M.: Wave propagation in continuous periodic structures: research contributions from Southampton, 1964–1995. *J. Sound Vib.* **190**, 495–524 (1996)
15. Ruzzene, M., Scarpa, F., Soranna, F.: Wave beaming effects in two-dimensional cellular structures. *Smart Mater. Struct.* **12**, 363 (2003)
16. Phani, A.S., Woodhouse, J., Fleck, N.A.: Wave propagation in two-dimensional periodic lattices. *J. Acoust. Soc. Am.* **119**, 1995–2005 (2006)
17. Gonella, S., Ruzzene, M.: Analysis of in-plane wave propagation in hexagonal and re-entrant lattices. *J. Sound Vib.* **312**, 125–139 (2008)
18. Farzbod, F., Leamy, M.J.: Analysis of Bloch's method and the propagation technique in periodic structures (2011)
19. Farzbod, F., Leamy, M.J.: The treatment of forces in Bloch analysis. *J. Sound Vib.* **325**, 545–551 (2009)
20. Gupta, A.K., Taufik, M.: Fabrication of Functionally Graded Materials: A Review. In: *International Conference on Production and Industrial Engineering*, pp. 79–89. Springer (2023)
21. Singh, A., Kumari, P., Sharma, D.: Axially Functionally Graded Beams—A Review. In: *Asian Conference on Mechanics of Functional Materials and Structures*, pp. 479–496. Springer (2022)
22. Meenu Krishnan, U., Gupta, A., Chowdhury, R.: Topology Optimization of Metamaterials Using Functionally Graded Material. In: *National Conference on Multidisciplinary Analysis and Optimization*, pp. 17–24. Springer (2025)
23. Mishra, M., Srivastav, C.K., Kumar, S.: Free vibration analysis using dynamic stiffness method and first-order shear deformation theory for a functionally graded material plate. *Mater Today Proc.* **98**, 7–15 (2024)
24. Huang, Z., Han, M., Wang, X., Chu, F.: Free vibration of functionally graded material sandwich plates with soft core. *J. Vib. Eng. Technol.* **12**, 5119–5131 (2024)
25. Van Doan, D., Van Minh, P., Van Ke, T., Nhung, N.T.C., Van Thom, D.: An overview of functionally graded materials: from civil applications to defense and aerospace industries. *J. Vib. Eng. Technol.* **13**, 1–20 (2025)
26. Sepehri, S., Jafari, H., Mosavi Mashhadi, M., Hairi Yazdi, M.R., Seyyed Fakhraabadi, M.M.: Tunable elastic wave propagation in planar functionally graded metamaterials. *Acta Mech.* **231**, 3363–3385 (2020)
27. Jafari, H., Sepehri, S., Yazdi, M.R.H., Mashhadi, M.M., Fakhraabadi, M.M.S.: Hybrid lattice metamaterials with auxiliary resonators made of functionally graded materials. *Acta Mech.* **231**, 4835–4849 (2020)
28. Li, J., Yang, P., Li, S.: Multiple band gaps for efficient wave attenuation by inertial amplification in periodic functionally graded beams. *Compos. Struct.* **271**, 114130 (2021)
29. Li, Y.S., Li, S.: Vibro-acoustic response and sound transmission loss of functionally graded graphene origami-enabled auxetic metamaterial plates. *Acta Mech.* **66**, 1–24s (2025)
30. Kumar, R., Jana, P.: Prestressed vibration and buckling analysis of graphene platelets reinforced composite plates using the dynamic stiffness method. *Acta Mech.* **235**, 5963–5992 (2024)
31. Zhu, H., Wu, J.: Free vibration of partially fluid-filled or fluid-surrounded composite shells using the dynamic stiffness method. *Acta Mech.* **231**, 3961–3978 (2020)
32. Liu, X., Lu, Z., Adhikari, S., Li, Y., Banerjee, J.R.: Exact wave propagation analysis of lattice structures based on the dynamic stiffness method and the Wittrick-Williams algorithm. *Mech. Syst. Signal Process.* **174**, 109044 (2022)
33. Neyaa, B.N., Nateghi-Babagi, P.: Free vibration of functionally graded thick transversely isotropic rectangular plates: an exact and three dimensional solution. *J. Vib. Eng. Technol.* **13**, 198 (2025)
34. Wu, M.-L., Wu, L.-Y., Yang, W.-P., Chen, L.-W.: Elastic wave band gaps of one-dimensional phononic crystals with functionally graded materials. *Smart Mater. Struct.* **18**, 115013 (2009)
35. Li, J., Li, S., Lin, W.: Actively controllable size-dependent elastic wave band gaps in planar functionally graded micro-lattice. *Mech. Adv. Mater. Struct.* **31**, 1678–1698 (2024)
36. Yan, L., Shi, D., He, D., Wang, L., Zhao, K.: Band gap characteristics of two-dimensional functionally graded periodic grid structures with local resonators. *Mech. Adv. Mater. Struct.* **30**, 4028–4039 (2023)
37. Gao, X.-Y., Wang, Z.-Z., Ma, L.-S.: Free vibration and buckling analysis of FG graphene origami-enabled auxetic metamaterial beams in a thermal environment. *Acta Mech.* **236**, 1265–1287 (2025)
38. Dorn, C., Kochmann, D.M.: Ray theory for elastic wave propagation in graded metamaterials. *J. Mech. Phys. Solids* **168**, 105049 (2022)
39. Bacigalupo, A., Badino, P., Gambarotta, L.: Dynamic homogenization of multi-layered lattice-like metamaterials with alternate chiral microstructure. *Meccanica* (2025). <https://doi.org/10.1007/s11012-025-01966-8>
40. Wittrick, W., Williams, F.: A general algorithm for computing natural frequencies of elastic structures. *Q. J. Mech. Appl. Math.* **24**, 263–284 (1971)
41. Mishra, M., Kumar, S.: Analysis of wave propagation in functionally graded square lattice structures using the Dynamic Stiffness Method combined with the Wittrick-Williams Algorithm. *Iran. J. Sci. Technol. Trans. Mech. Eng.* (2025). <https://doi.org/10.1007/s40997-025-00883-1>
42. Birman, V., Byrd, L.W.: Modeling and analysis of functionally graded materials and structures (2007)
43. Chi, S.-H., Chung, Y.-L.: Mechanical behavior of functionally graded material plates under transverse load—Part I: analysis. *Int. J. Solids Struct.* **43**, 3657–3674 (2006)
44. Banerjee, J.R., Ananthapuvirajah, A.: An exact dynamic stiffness matrix for a beam incorporating Rayleigh–Love and Timoshenko theories. *Int. J. Mech. Sci.* **150**, 337–347 (2019)
45. Liu, X., Zhao, Y., Zhou, W., Banerjee, J.R.: Dynamic stiffness method for exact longitudinal free vibration of rods and trusses using simple and advanced theories. *Appl. Math. Model.* **104**, 401–420 (2022)
46. Zhang, K., Zhao, P., Zhao, C., Hong, F., Deng, Z.: Study on the mechanism of band gap and directional wave propagation of the auxetic chiral lattices. *Compos. Struct.* **238**, 111952 (2020)

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